Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Nyman Brothers Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
20445 August AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1907

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0015-0001-0037-0000 0045-0014-0018-0000

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: (209) 495-8821 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This operator is responsible for paying permit fees.

Nyman, Lloyd

Operator name: Nyman, Lloyd

Telephone no.: (209) 634-1046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
20445 August AVE	Hilmar	CA	95324

C. OWNERS

Nyman, Brad

Legal owner name: Nyman, Brad

Telephone no.: (209) 495-8821
Landline | Cellular

P.O. Box 1331 | Hilmar | CA | 95324
Mailing Address Number and Street | City | State | Zip Code

This owner is responsible for paying permit fees.

Nyman, Lloyd

Legal owner name: Nyman, Lloyd

Telephone no.: (209) 634-1046
Landline | Cellular

20445 August AVE | Hilmar | CA | 95324
Mailing Address Number and Street | City | State | Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	250	250	0	500	250
Number under roof	1,025	0	0	0	0	250
Maximum number	1,075	250	250	0	500	500
Average number	1,025	250	250	0	500	500
Avg live weight (lbs)	950	1,000	700	0		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 34,117.95 *tons per reporting period*

Total nitrogen from manure: 448,486.39 *lbs per reporting period*

After ammonia losses (30% loss applied): 313,940.47 *lbs per reporting period*

Total phosphorus from manure: 68,956.29 *lbs per reporting period*

Total potassium from manure: 189,562.71 *lbs per reporting period*

Total salt from manure: 540,108.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 38,157,500 *gallons*

Total nitrogen generated: 184,326.15 *lbs*

Total phosphorus generated: 21,206.56 *lbs*

Total potassium generated: 289,732.48 *lbs*

Total salt generated: 1,511,347.51 *lbs*

	30,575,000	*gallons applied*
+	7,582,500	*gallons exported*
-	0	*gallons imported*
=	38,157,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	Corral solids	1,890.00 *ton*	Dry-weight	39.6		16,100.00	2,260.00	3,530.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
02/26/2023	Process wastewater	1,755,000.00 *gal*	714.00	339.00	0.00	0.00	101.00	640.00		4,260
05/16/2023	Process wastewater	1,485,000.00 *gal*	554.00	384.00	0.00	0.00	48.60	1,090.00		4,340
07/13/2023	Process wastewater	3,240,000.00 *gal*	554.00	384.00	0.00	0.00	48.60	1,090.00		4,340
08/11/2023	Process wastewater	1,102,500.00 *gal*	168.00	104.00	0.00	0.00	60.10	713.00		3,900

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	36,758.23	5,159.85	8,059.41	0.00
Process wastewater	33,846.82	3,948.44	58,911.80	269,397.88
Total exports for all materials	70,605.05	9,108.29	66,971.21	269,397.88

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
August 1	30	30	2	process wastewater	X045-X017-X052-XXXX X045-X017-X053-XXXX X045-X017-X055-XXXX
August 2	35	35	2	process wastewater	X045-X017-X045-XXXX
August 3	9	9	2	process wastewater	X015-X010-X037-XXXX
August 4	28	28	2	process wastewater	X015-X010-X003-XXXX
Lander 1	32	32	2	process wastewater	X015-X009-X001-XXXX X015-X009-X045-XXXX
Lander 10	18	18	1	process wastewater	X045-X016-X089-XXXX X045-X016-X090-XXXX
Lander 11	17	17	1	process wastewater	X045-X016-X006-XXXX
Lander 2 & 3	17	17	2	process wastewater	X015-X009-X054-XXXX X015-X009-X056-XXXX
Lander 4 & 6	20	20	1	process wastewater	X045-X016-X002-XXXX X045-X016-X074-XXXX X045-X016-X090-XXXX
Lander 5 & 7	25	25	1	process wastewater	X015-X009-X054-XXXX X045-X016-X002-XXXX X045-X016-X074-XXXX X045-X016-X090-XXXX
Lander 8	26	26	1	process wastewater	X045-X160-X074-XXXX
Lander 9	4	4	1	process wastewater	X045-X016-X074-XXXX
Oslo 1	10	10	2	process wastewater	X045-X014-X030-XXXX
Oslo 2	38	38	2	process wastewater	X045-X014-X030-XXXX
Oslo 3	16	16	2	process wastewater	X045-X014-X030-XXXX
Oslo 4	24	24	2	process wastewater	X045-X014-X018-XXXX X045-X014-X041-XXXX
Oslo 5	11	11	2	process wastewater	X045-X140-X087-XXXX
Verde	25	16	2	process wastewater	X045-X036-X014-XXXX
Totals for areas that were used for application	385	376	30		
Totals for areas that were not used for application					
Land application area totals	385	376	30		

B. CROPS AND HARVESTS

August 1

Field name: August 1

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	560.00 *ton*	As-is		67.4	6,290.00	2,090.00	26,100.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	18.67	234.83	78.03	974.40	1,533.50

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	890.00 *ton*	As-is		63.0	4,270.00	1,110.00	6,290.00		3.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	29.67	253.35	65.86	373.21	875.94

August 2

Field name: August 2

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	590.00 *ton*	As-is		66.7	8,390.00	1,960.00	22,600.00		9.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	16.86	282.86	66.08	761.94	1,109.21

August 2

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,054.00 *ton*	As-is		58.0	4,390.00	1,220.00	8,400.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	30.11	264.40	73.48	505.92	1,214.21

August 3

Field name: August 3

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	155.00 *ton*	As-is		74.5	6,480.00	1,480.00	18,100.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	17.22	223.20	50.98	623.44	974.95

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	245.00 *ton*	As-is		62.4	4,220.00	1,240.00	10,200.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.22	229.76	67.51	555.33	1,228.27

August 4

Field name: August 4

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	515.00 *ton*	As-is		66.2	7,670.00	1,620.00	18,900.00		9.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.39	282.15	59.59	695.25	1,184.92

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	795.00 *ton*	As-is		57.6	5,200.00	1,360.00	11,400.00		6.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	28.39	295.29	77.23	647.36	1,562.61

Lander 1

Field name: Lander 1

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	520.00 *ton*	As-is		67.6	4,960.00	1,460.00	15,900.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.25	161.20	47.45	516.75	947.70

Lander 1

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	793.00 *ton*	As-is		58.8	3,980.00	1,360.00	9,060.00		5.81

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.78	197.26	67.41	449.04	1,186.39

Lander 10

Field name: Lander 10

01/01/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 18 Plant date: 01/01/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	126.00 *ton*	As-is		8.3	29,300.00	2,380.00	32,100.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	240.00	21.60	176.00	0.00
Total actual harvest content	7.00	410.20	33.32	449.40	1,450.69

Lander 11

Field name: Lander 11

Lander 11

01/01/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 17 Plant date: 01/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	117.00 *ton*	As-is		12.5	27,900.00	2,190.00	31,500.00		13.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	308.00	0.00
Total actual harvest content	6.88	384.04	30.14	433.59	1,613.91

Lander 2 & 3

Field name: Lander 2 & 3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	240.00 *ton*	As-is		69.6	6,260.00	1,950.00	22,200.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.12	176.75	55.06	626.82	909.85

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	405.00 *ton*	As-is		64.8	3,900.00	1,060.00	8,450.00		5.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	23.82	185.82	50.51	402.62	957.67

Lander 4 & 6

Field name: Lander 4 & 6

01/01/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 20 Plant date: 01/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	125.00 *ton*	As-is		7.0	30,000.00	2,420.00	30,700.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	308.00	0.00
Total actual harvest content	6.25	375.00	30.25	383.75	1,604.25

Lander 5 & 7

Field name: Lander 5 & 7

01/01/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 25 Plant date: 01/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	140.00 *ton*	As-is		7.7	29,700.00	2,860.00	40,600.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	308.00	0.00
Total actual harvest content	5.60	332.64	32.03	454.72	1,240.51

Lander 8

Field name: Lander 8

Lander 8

01/01/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 26 Plant date: 01/01/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	175.00 *ton*	As-is		8.8	29,200.00	2,550.00	33,700.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	240.00	21.60	176.00	0.00
Total actual harvest content	6.73	393.08	34.33	453.65	1,362.74

Lander 9

Field name: Lander 9

01/01/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 4 Plant date: 01/01/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	24.00 *ton*	As-is		8.3	29,300.00	2,380.00	32,100.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	240.00	21.60	176.00	0.00
Total actual harvest content	6.00	351.60	28.56	385.20	1,243.45

Oslo 1

Field name: Oslo 1

Oslo 1

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	140.00 *ton*	As-is		67.7	6,070.00	1,970.00	17,800.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.00	169.96	55.16	498.40	958.66

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	290.00 *ton*	As-is		55.0	4,190.00	1,260.00	9,900.00		5.21

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	29.00	243.02	73.08	574.20	1,359.81

Oslo 2

Field name: Oslo 2

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	540.00 *ton*	As-is		63.4	4,900.00	1,570.00	16,800.00		9.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.21	139.26	44.62	477.47	942.43

Oslo 2

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,104.00 *ton*	As-is		63.8	3,450.00	1,050.00	7,240.00		4.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	29.05	200.46	61.01	420.68	1,049.60

Oslo 3

Field name: Oslo 3

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	180.00 *ton*	As-is		67.6	7,940.00	1,850.00	21,100.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.25	178.65	41.63	474.75	809.19

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	465.00 *ton*	As-is		57.8	4,830.00	1,140.00	7,600.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	29.06	280.74	66.26	441.75	1,201.91

Oslo 4

Field name: Oslo 4

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	370.00 *ton*	As-is		63.2	4,860.00	1,550.00	17,700.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.42	149.85	47.79	545.75	907.73

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	730.00 *ton*	As-is		56.2	4,450.00	1,490.00	10,500.00		5.41

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	30.42	270.71	90.64	638.75	1,441.49

Oslo 5

Field name: Oslo 5

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	170.00 *ton*	As-is		63.8	4,960.00	1,340.00	15,900.00		8.72

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	15.45	153.31	41.42	491.45	975.69

Oslo 5

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	318.00 *ton*	As-is		60.6	4,420.00	1,100.00	7,490.00		5.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	28.91	255.56	63.60	433.06	1,318.98

Verde

Field name: Verde

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	180.00 *ton*	As-is		71.0	5,250.00	1,540.00	21,200.00		13.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	11.25	118.13	34.65	477.00	874.35

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	434.00 *ton*	As-is		57.6	4,570.00	1,230.00	8,480.00		4.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.13	247.92	66.73	460.04	1,101.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

August 1 - 11/02/2022: Oats, silage-soft dough

Field name: August 1

Crop: Oats, silage-soft dough — Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.56	9.12	136.24	1,096.53	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	237.26	4,374,000.00 *gal*
Application event totals		81.56	9.12	136.24	1,333.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.37	12.64	80.11	533.25	450,000.00 *gal*
Application event totals		89.37	12.64	80.11	533.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.50	10.11	64.09	426.60	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	158.17	2,916,000.00 *gal*
Application event totals		71.50	10.11	64.09	584.77	

August 1 - 05/22/2023: Corn, silage

Field name: August 1

Crop: Corn, silage — Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

August 1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.22	7.30	163.73	651.91	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	237.26	4,374,000.00 *gal*
Application event totals		83.22	7.30	163.73	889.17	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	257.03	4,738,500.00 *gal*
Application event totals		0.00	0.00	0.00	257.03	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	217.49	4,009,500.00 *gal*
Application event totals		0.00	0.00	0.00	217.49	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	197.71	3,645,000.00 *gal*
Application event totals		69.35	6.08	136.44	740.97	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

August 1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.41	5.48	122.80	488.93	405,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	177.94	3,280,500.00 *gal*
Application event totals		62.41	5.48	122.80	666.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	197.71	3,645,000.00 *gal*
Application event totals		69.35	6.08	136.44	740.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.93	6.77	80.32	439.36	405,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	177.94	3,280,500.00 *gal*
Application event totals		18.93	6.77	80.32	617.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	158.17	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	158.17	

August 2 - 11/02/2022: Oats, silage-soft dough

Field name: August 2

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

August 2 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.74	8.47	126.51	1,018.21	585,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	220.31	4,738,500.00 *gal*
Application event totals		75.74	8.47	126.51	1,238.52	

01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.61	10.84	68.67	457.07	450,000.00 *gal*
Application event totals		76.61	10.84	68.67	457.07	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.93	13.00	82.40	548.48	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	203.36	4,374,000.00 *gal*
Application event totals		91.93	13.00	82.40	751.84	

August 2 - 05/22/2023: Corn, silage

Field name: August 2

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.16	7.82	175.42	698.48	675,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	271.15	5,832,000.00 *gal*
Application event totals		89.16	7.82	175.42	969.63	

August 2 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	254.20	5,467,500.00 *gal*
Application event totals		0.00	0.00	0.00	254.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	186.42	4,009,500.00 *gal*
Application event totals		0.00	0.00	0.00	186.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.44	5.21	116.95	465.65	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	237.26	5,103,000.00 *gal*
Application event totals		59.44	5.21	116.95	702.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	169.47	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	169.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.50	4.69	105.25	419.09	405,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	169.47	3,645,000.00 *gal*
Application event totals		53.50	4.69	105.25	588.55	

August 2 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.44	5.21	116.95	465.65	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	186.42	4,009,500.00 *gal*
Application event totals		59.44	5.21	116.95	652.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.22	5.80	68.85	376.60	405,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	203.36	4,374,000.00 *gal*
Application event totals		16.22	5.80	68.85	579.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	186.42	4,009,500.00 *gal*
Application event totals		0.00	0.00	0.00	186.42	

August 3 - 11/02/2022: Oats, silage-soft dough

Field name: August 3

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.63	10.13	151.38	1,218.37	180,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	263.62	1,458,000.00 *gal*
Application event totals		90.63	10.13	151.38	1,481.99	

August 3 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.37	12.64	80.11	533.25	135,000.00 *gal*
Application event totals		89.37	12.64	80.11	533.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.17	16.86	106.82	710.99	180,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	263.62	1,458,000.00 *gal*
Application event totals		119.17	16.86	106.82	974.61	

August 3 - 05/22/2023: Corn, silage

Field name: August 3

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.90	7.10	159.18	633.80	157,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	230.67	1,275,750.00 *gal*
Application event totals		80.90	7.10	159.18	864.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	263.62	1,458,000.00 *gal*
Application event totals		0.00	0.00	0.00	263.62	

August 3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	164.76	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	164.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	135,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	197.71	1,093,500.00 *gal*
Application event totals		69.35	6.08	136.44	740.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	197.71	1,093,500.00 *gal*
Application event totals		0.00	0.00	0.00	197.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.79	5.07	113.70	452.72	112,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	164.76	911,250.00 *gal*
Application event totals		57.79	5.07	113.70	617.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.79	5.07	113.70	452.72	112,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	164.76	911,250.00 *gal*
Application event totals		57.79	5.07	113.70	617.48	

August 3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.02	5.02	59.50	325.46	90,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	131.81	729,000.00 *gal*
Application event totals		14.02	5.02	59.50	457.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	197.71	1,093,500.00 *gal*
Application event totals		0.00	0.00	0.00	197.71	

August 4 - 11/02/2022: Oats, silage-soft dough

Field name: August 4

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.39	9.77	145.97	1,174.86	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	254.20	4,374,000.00 *gal*
Application event totals		87.39	9.77	145.97	1,429.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.76	13.55	85.83	571.33	450,000.00 *gal*
Application event totals		95.76	13.55	85.83	571.33	

August 4 - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.61	10.84	68.67	457.07	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	169.47	2,916,000.00 *gal*
Application event totals		76.61	10.84	68.67	626.54	

August 4 - 05/22/2023: Corn, silage

Field name: August 4

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.16	7.82	175.42	698.48	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	254.20	4,374,000.00 *gal*
Application event totals		89.16	7.82	175.42	952.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	254.20	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	254.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	190.65	3,280,500.00 *gal*
Application event totals		0.00	0.00	0.00	190.65	

August 4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.73	7.17	160.81	640.27	495,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	233.02	4,009,500.00 *gal*
Application event totals		81.73	7.17	160.81	873.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	190.65	3,280,500.00 *gal*
Application event totals		0.00	0.00	0.00	190.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.87	5.87	131.57	523.86	405,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	190.65	3,280,500.00 *gal*
Application event totals		66.87	5.87	131.57	714.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.30	6.52	146.19	582.06	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	211.84	3,645,000.00 *gal*
Application event totals		74.30	6.52	146.19	793.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.90	6.05	71.72	392.29	337,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	158.88	2,733,750.00 *gal*
Application event totals		16.90	6.05	71.72	551.17	

August 4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	158.88	2,733,750.00 *gal*
Application event totals		0.00	0.00	0.00	158.88	

Lander 1 - 11/01/2022: Oats, silage-soft dough

Field name: Lander 1

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.96	11.40	170.30	1,370.67	720,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	296.57	5,832,000.00 *gal*
Application event totals		101.96	11.40	170.30	1,667.24	

02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.55	14.22	90.13	599.90	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	222.43	4,374,000.00 *gal*
Application event totals		100.55	14.22	90.13	822.33	

Lander 1 - 05/24/2023: Corn, silage

Field name: Lander 1

Crop: Corn, silage Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Lander 1 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.02	7.98	179.08	713.03	630,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	259.50	5,103,000.00 *gal*
Application event totals		91.02	7.98	179.08	972.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.51	5.13	115.12	458.38	405,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	166.82	3,280,500.00 *gal*
Application event totals		58.51	5.13	115.12	625.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	185.36	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	185.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	203.89	4,009,500.00 *gal*
Application event totals		0.00	0.00	0.00	203.89	

Lander 1 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	148.29	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	148.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.01	5.70	127.91	509.31	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	185.36	3,645,000.00 *gal*
Application event totals		65.01	5.70	127.91	694.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	166.82	3,280,500.00 *gal*
Application event totals		0.00	0.00	0.00	166.82	

Lander 10 - 01/01/2021: Alfalfa, hay

Field name: Lander 10

Crop: Alfalfa, hay

Plant date: 01/01/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	148.96	21.07	133.52	888.74	450,000.00 *gal*
Application event totals		148.96	21.07	133.52	888.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	296.57	3,280,500.00 *gal*
Application event totals		0.00	0.00	0.00	296.57	

Lander 10 - 01/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	263.62	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	263.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	263.62	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	263.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	230.67	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	230.67	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	230.67	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	230.67	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

Lander 10 - 01/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	230.67	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	230.67	

Lander 11 - 01/01/2018: Alfalfa, hay

Field name: Lander 11

Crop: Alfalfa, hay

Plant date: 01/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	157.72	22.31	141.37	941.02	450,000.00 *gal*
Application event totals		157.72	22.31	141.37	941.02	

03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	314.02	3,280,500.00 *gal*
Application event totals		0.00	0.00	0.00	314.02	

04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Lander 11 - 01/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	2,442.01	25,511,500.00 *gal*
Application event totals		0.00	0.00	0.00	2,442.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Lander 2 & 3 - 11/01/2022: Oats, silage-soft dough

Field name: Lander 2 & 3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Lander 2 & 3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.96	10.73	160.28	1,290.04	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	279.13	2,916,000.00 *gal*
Application event totals		95.96	10.73	160.28	1,569.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.63	13.39	84.82	564.61	270,000.00 *gal*
Application event totals		94.63	13.39	84.82	564.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Lander 2 & 3 - 05/24/2023: Corn, silage

Field name: Lander 2 & 3

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.66	7.51	168.54	671.09	315,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		85.66	7.51	168.54	915.32	

Lander 2 & 3 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	244.23	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	244.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	209.34	2,187,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.19	5.37	120.39	479.35	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	174.45	1,822,500.00 *gal*
Application event totals		61.19	5.37	120.39	653.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	226.79	2,369,250.00 *gal*
Application event totals		0.00	0.00	0.00	226.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	174.45	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	174.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	139.56	1,458,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.56	

Lander 2 & 3 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.19	5.37	120.39	479.35	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	174.45	1,822,500.00 *gal*
Application event totals		61.19	5.37	120.39	653.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	174.45	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	174.45	

Lander 4 & 6 - 01/01/2018: Alfalfa, hay

Field name: Lander 4 & 6

Crop: Alfalfa, hay

Plant date: 01/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	160.87	22.76	144.20	959.84	540,000.00 *gal*
Application event totals		160.87	22.76	144.20	959.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	355.89	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	355.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Lander 4 & 6 - 01/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Lander 5 & 7 - 01/01/2018: Alfalfa, hay

Lander 5 & 7 - 01/01/2018: Alfalfa, hay

Field name: Lander 5 & 7

Crop: Alfalfa, hay

Plant date: 01/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	150.15	21.24	134.59	895.85	630,000.00 *gal*
Application event totals		150.15	21.24	134.59	895.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	284.71	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	284.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Lander 5 & 7 - 01/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	189.81	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.81	

Lander 8 - 01/01/2021: Alfalfa, hay

Field name: Lander 8

Crop: Alfalfa, hay

Plant date: 01/01/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	154.69	21.88	138.66	922.92	675,000.00 *gal*
Application event totals		154.69	21.88	138.66	922.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	285.16	4,556,250.00 *gal*
Application event totals		0.00	0.00	0.00	285.16	

Lander 8 - 01/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Lander 8 - 01/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	228.13	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.13	

Lander 9 - 01/01/2021: Alfalfa, hay

Field name: Lander 9

Crop: Alfalfa, hay

Plant date: 01/01/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	167.58	23.71	150.21	999.84	112,500.00 *gal*
Application event totals		167.58	23.71	150.21	999.84	

03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	296.57	729,000.00 *gal*
Application event totals		0.00	0.00	0.00	296.57	

04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Lander 9 - 01/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	546,750.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Oslo 1 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 1

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Oslo 1 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.96	11.40	170.30	1,370.67	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	237.26	1,458,000.00 *gal*
Application event totals		101.96	11.40	170.30	1,607.92	

02/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.06	18.96	120.17	799.87	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	296.57	1,822,500.00 *gal*
Application event totals		134.06	18.96	120.17	1,096.44	

Oslo 1 - 05/23/2023: Corn, silage

Field name: Oslo 1

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.02	9.13	204.66	814.89	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	296.57	1,822,500.00 *gal*
Application event totals		104.02	9.13	204.66	1,111.46	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	1,458,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Oslo 1 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.41	5.48	122.80	488.93	135,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	177.94	1,093,500.00 *gal*
Application event totals		62.41	5.48	122.80	666.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	237.26	1,458,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.01	4.56	102.33	407.44	112,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	148.29	911,250.00 *gal*
Application event totals		52.01	4.56	102.33	555.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	207.60	1,275,750.00 *gal*
Application event totals		0.00	0.00	0.00	207.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.41	5.48	122.80	488.93	135,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	177.94	1,093,500.00 *gal*
Application event totals		62.41	5.48	122.80	666.88	

Oslo 1 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	177.94	1,093,500.00 *gal*
Application event totals		0.00	0.00	0.00	177.94	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	148.29	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	148.29	

Oslo 2 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 2

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.86	9.60	143.41	1,154.25	720,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	249.74	5,832,000.00 *gal*
Application event totals		85.86	9.60	143.41	1,403.99	

02/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.84	14.97	94.87	631.47	675,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	234.13	5,467,500.00 *gal*
Application event totals		105.84	14.97	94.87	865.61	

Oslo 2 - 05/23/2023: Corn, silage

Oslo 2 - 05/23/2023: Corn, silage

Field name: Oslo 2

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.60	7.68	172.35	686.22	720,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	249.74	5,832,000.00 *gal*
Application event totals		87.60	7.68	172.35	935.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	249.74	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	249.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.70	5.76	129.26	514.67	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	187.31	4,374,000.00 *gal*
Application event totals		65.70	5.76	129.26	701.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	218.53	5,103,000.00 *gal*
Application event totals		0.00	0.00	0.00	218.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.70	5.76	129.26	514.67	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	187.31	4,374,000.00 *gal*
Application event totals		65.70	5.76	129.26	701.97	

Oslo 2 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	234.13	5,467,500.00 *gal*
Application event totals		0.00	0.00	0.00	234.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.22	5.28	118.49	471.78	495,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	171.70	4,009,500.00 *gal*
Application event totals		60.22	5.28	118.49	643.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	187.31	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	187.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	187.31	4,374,000.00 *gal*
Application event totals		0.00	0.00	0.00	187.31	

Oslo 3 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 3

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Oslo 3 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.58	10.68	159.66	1,285.00	337,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	278.04	2,733,750.00 *gal*
Application event totals		95.58	10.68	159.66	1,563.03	

02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.55	14.22	90.13	599.90	270,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	222.43	2,187,000.00 *gal*
Application event totals		100.55	14.22	90.13	822.33	

Oslo 3 - 05/23/2023: Corn, silage

Field name: Oslo 3

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.02	9.13	204.66	814.89	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	296.57	2,916,000.00 *gal*
Application event totals		104.02	9.13	204.66	1,111.46	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	259.50	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.50	

Oslo 3 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.01	5.70	127.91	509.31	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	185.36	1,822,500.00 *gal*
Application event totals		65.01	5.70	127.91	694.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	2,187,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	166.82	1,640,250.00 *gal*
Application event totals		0.00	0.00	0.00	166.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.51	6.27	140.70	560.24	247,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	203.89	2,004,750.00 *gal*
Application event totals		71.51	6.27	140.70	764.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.01	5.70	127.91	509.31	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	185.36	1,822,500.00 *gal*
Application event totals		65.01	5.70	127.91	694.66	

Oslo 3 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	185.36	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	185.36	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	148.29	1,458,000.00 *gal*
Application event totals		0.00	0.00	0.00	148.29	

Oslo 4 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 4

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.96	11.40	170.30	1,370.67	540,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	296.57	4,374,000.00 *gal*
Application event totals		101.96	11.40	170.30	1,667.24	

02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.72	15.80	100.14	666.56	450,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	247.14	3,645,000.00 *gal*
Application event totals		111.72	15.80	100.14	913.70	

Oslo 4 - 05/23/2023: Corn, silage

Oslo 4 - 05/23/2023: Corn, silage

Field name: Oslo 4

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.69	9.89	221.72	882.80	585,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	321.29	4,738,500.00 *gal*
Application event totals		112.69	9.89	221.72	1,204.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	247.14	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	247.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	197.71	2,916,000.00 *gal*
Application event totals		69.35	6.08	136.44	740.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	185.36	2,733,750.00 *gal*
Application event totals		0.00	0.00	0.00	185.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	148.29	2,187,000.00 *gal*
Application event totals		0.00	0.00	0.00	148.29	

Oslo 4 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.35	6.08	136.44	543.26	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	197.71	2,916,000.00 *gal*
Application event totals		69.35	6.08	136.44	740.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.68	5.32	119.39	475.35	315,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	173.00	2,551,500.00 *gal*
Application event totals		60.68	5.32	119.39	648.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	173.00	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	173.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	148.29	2,187,000.00 *gal*
Application event totals		0.00	0.00	0.00	148.29	

Oslo 5 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 5

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Oslo 5 - 11/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.42	9.32	139.34	1,121.45	202,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	269.61	1,822,500.00 *gal*
Application event totals		83.42	9.32	139.34	1,391.06	

02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.50	13.79	87.39	581.72	180,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	215.69	1,458,000.00 *gal*
Application event totals		97.50	13.79	87.39	797.41	

Oslo 5 - 05/23/2023: Corn, silage

Field name: Oslo 5

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.11	7.47	167.45	666.73	202,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	242.65	1,640,250.00 *gal*
Application event totals		85.11	7.47	167.45	909.38	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	269.61	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	269.61	

Oslo 5 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.63	6.42	94.63	764.71	157,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	188.73	1,275,750.00 *gal*
Application event totals		67.63	6.42	94.63	953.43	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	242.65	1,640,250.00 *gal*
Application event totals		0.00	0.00	0.00	242.65	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	215.69	1,458,000.00 *gal*
Application event totals		0.00	0.00	0.00	215.69	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.61	9.17	135.19	1,092.44	225,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	269.61	1,822,500.00 *gal*
Application event totals		96.61	9.17	135.19	1,362.05	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.63	6.42	94.63	764.71	157,500.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	188.73	1,275,750.00 *gal*
Application event totals		67.63	6.42	94.63	953.43	

Oslo 5 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	161.77	1,093,500.00 *gal*
Application event totals		0.00	0.00	0.00	161.77	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	161.77	1,093,500.00 *gal*
Application event totals		0.00	0.00	0.00	161.77	

Verde - 11/04/2022: Oats, silage-soft dough

Field name: Verde

Crop: Oats, silage-soft dough Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	150.04	21.53	136.36	1,004.74	320,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	222.43	2,187,000.00 *gal*
Application event totals		150.04	21.53	136.36	1,227.17	

10/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	2,187,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

Verde - 06/12/2023: Corn, silage

Field name: Verde

Crop: Corn, silage Plant date: 06/12/2023

Verde - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.88	17.71	146.45	1,040.20	360,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	296.57	2,916,000.00 *gal*
Application event totals		139.88	17.71	146.45	1,336.78	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.20	12.05	99.67	707.92	245,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	259.50	2,551,500.00 *gal*
Application event totals		95.20	12.05	99.67	967.42	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.97	13.03	107.81	765.71	265,000.00 *gal*
TID Canal	Surface water	0.00	0.00	0.00	222.43	2,187,000.00 *gal*
Application event totals		102.97	13.03	107.81	988.13	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	222.43	2,187,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.43	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	259.50	2,551,500.00 *gal*
Application event totals		0.00	0.00	0.00	259.50	

Verde - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	203.89	2,004,750.00 *gal*
Application event totals		0.00	0.00	0.00	203.89	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	185.36	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	185.36	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	185.36	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	185.36	

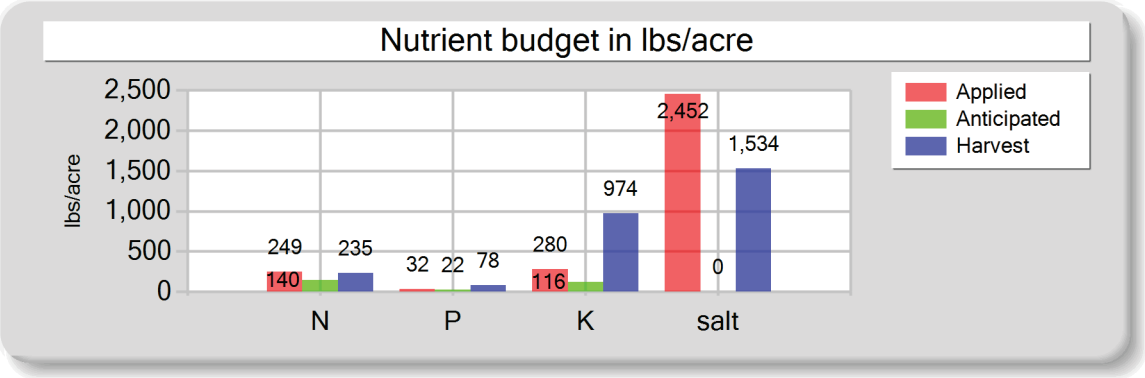
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Canal	Surface water	0.00	0.00	0.00	185.36	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	185.36	

B. NUTRIENT BUDGET

August 1 - 11/02/2022: Oats, silage-soft dough

Field name: August 1 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	242.44	31.87	280.44	2,056.37
Fresh water	0.00	0.00	0.00	395.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.44	31.87	280.44	2,451.80
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	234.83	78.03	974.40	1,533.50
Nutrient balance	14.61	-46.15	-693.96	918.30
Applied to removed ratio	1.06	0.41	0.29	1.60

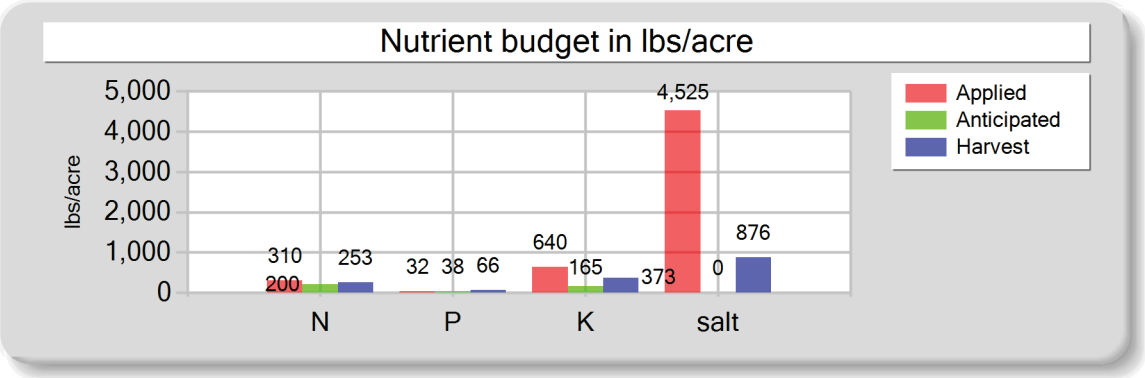
Fresh water applied
7,290,000.00 *gallons*
268.47 *acre-inches*
8.95 *inches/acre*

Process wastewater applied
1,350,000.00 *gallons*
49.72 *acre-inches*
1.66 *inches/acre*

Total harvests for the crop
1 *harvests*

August 1 - 05/22/2023: Corn, silage

Field name: August 1 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	303.25	31.71	639.73	2,666.73
Fresh water	0.00	0.00	0.00	1,858.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.25	31.71	639.73	4,525.24
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	253.35	65.86	373.21	875.94
Nutrient balance	56.90	-34.15	266.53	3,649.30
Applied to removed ratio	1.22	0.48	1.71	5.17

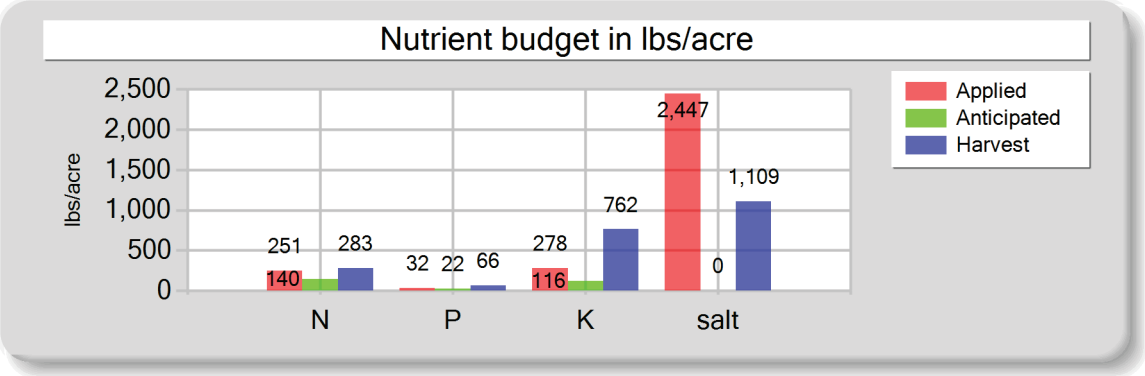
Fresh water applied	
34,263,000.00	*gallons*
1,261.79	*acre-inches*
42.06	*inches/acre*

Process wastewater applied	
2,250,000.00	*gallons*
82.86	*acre-inches*
2.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

August 2 - 11/02/2022: Oats, silage-soft dough

Field name: August 2 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	244.27	32.31	277.58	2,023.76
Fresh water	0.00	0.00	0.00	423.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.27	32.31	277.58	2,447.43
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	282.86	66.08	761.94	1,109.21
Nutrient balance	-31.59	-33.77	-484.37	1,338.22
Applied to removed ratio	0.89	0.49	0.36	2.21

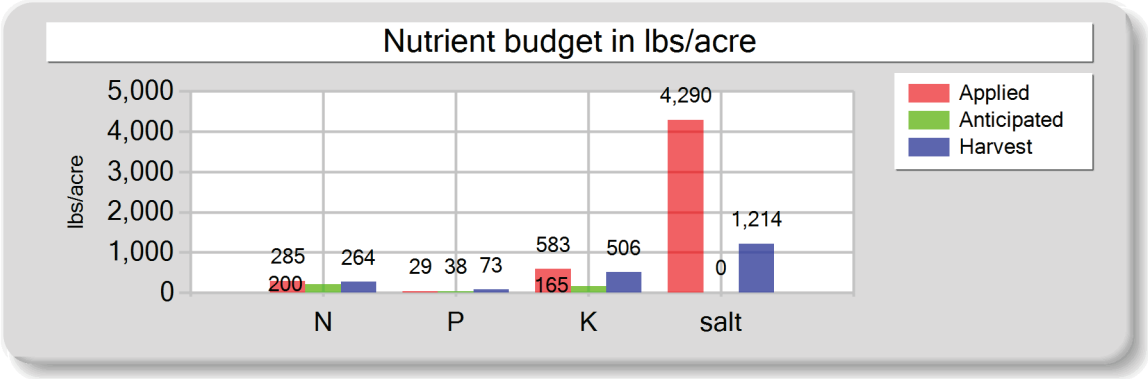
Fresh water applied	
9,112,500.00	*gallons*
335.58	*acre-inches*
9.59	*inches/acre*

Process wastewater applied	
1,575,000.00	*gallons*
58.00	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

August 2 - 05/22/2023: Corn, silage

Field name: August 2 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	277.76	28.75	583.43	2,425.46
Fresh water	0.00	0.00	0.00	1,864.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.76	28.75	583.43	4,289.62
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	264.40	73.48	505.92	1,214.21
Nutrient balance	20.36	-44.73	77.51	3,075.41
Applied to removed ratio	1.08	0.39	1.15	3.53

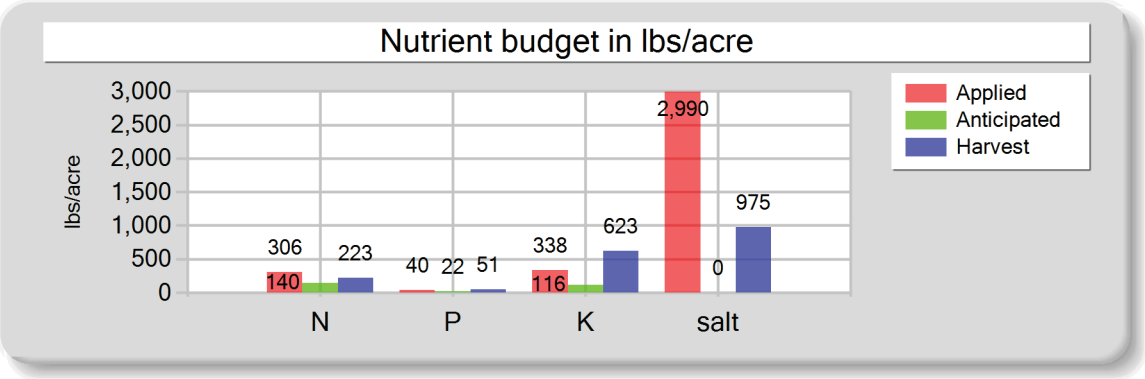
Fresh water applied	
40,095,000.00	*gallons*
1,476.56	*acre-inches*
42.19	*inches/acre*

Process wastewater applied	
2,385,000.00	*gallons*
87.83	*acre-inches*
2.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

August 3 - 11/02/2022: Oats, silage-soft dough

Field name: August 3 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	299.17	39.63	338.31	2,462.61
Fresh water	0.00	0.00	0.00	527.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.17	39.63	338.31	2,989.85
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	223.20	50.98	623.44	974.95
Nutrient balance	82.97	-11.35	-285.14	2,014.90
Applied to removed ratio	1.37	0.78	0.54	3.07

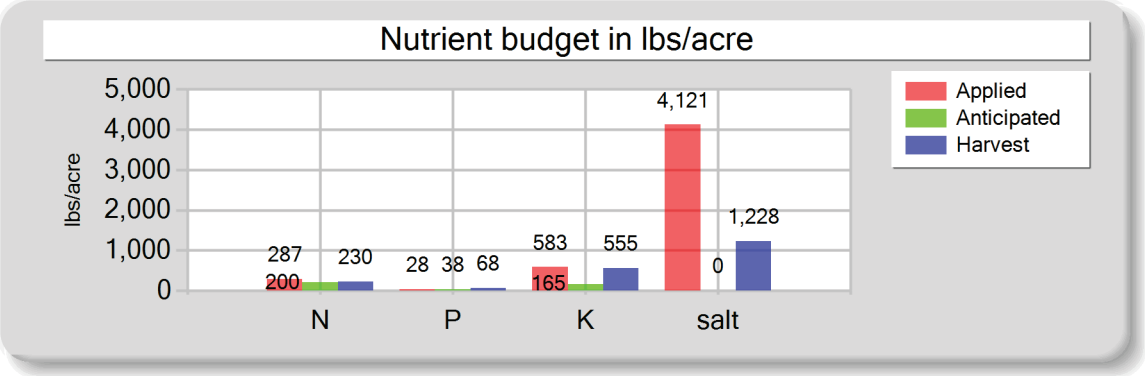
Fresh water applied	
2,916,000.00	*gallons*
107.39	*acre-inches*
11.93	*inches/acre*

Process wastewater applied	
495,000.00	*gallons*
18.23	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

August 3 - 05/22/2023: Corn, silage

Field name: August 3 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	279.85	28.34	582.52	2,407.95
Fresh water	0.00	0.00	0.00	1,713.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.85	28.34	582.52	4,121.47
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	229.76	67.51	555.33	1,228.27
Nutrient balance	57.09	-39.18	27.19	2,893.20
Applied to removed ratio	1.25	0.42	1.05	3.36

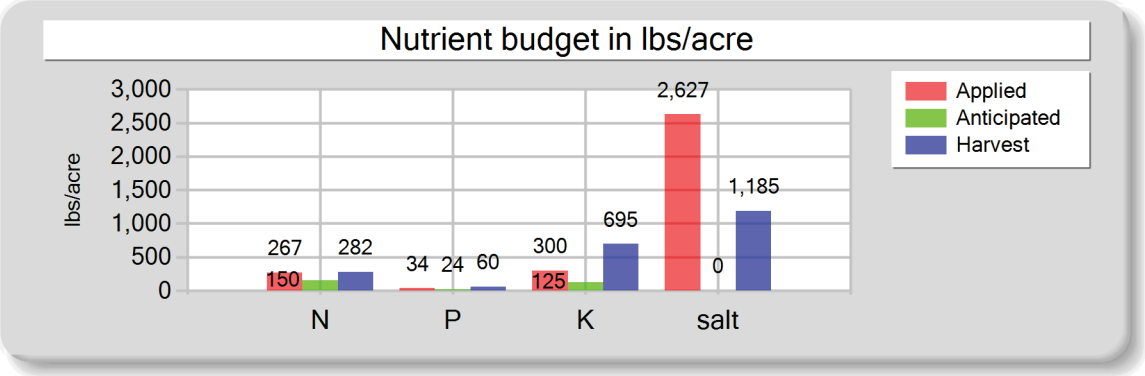
Fresh water applied	
9,477,000.00	*gallons*
349.01	*acre-inches*
38.78	*inches/acre*

Process wastewater applied	
607,500.00	*gallons*
22.37	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

August 4 - 11/02/2022: Oats, silage-soft dough

Field name: August 4 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	259.76	34.15	300.47	2,203.26
Fresh water	0.00	0.00	0.00	423.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.76	34.15	300.47	2,626.93
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	282.15	59.59	695.25	1,184.92
Nutrient balance	-15.39	-25.44	-394.78	1,442.01
Applied to removed ratio	0.95	0.57	0.43	2.22

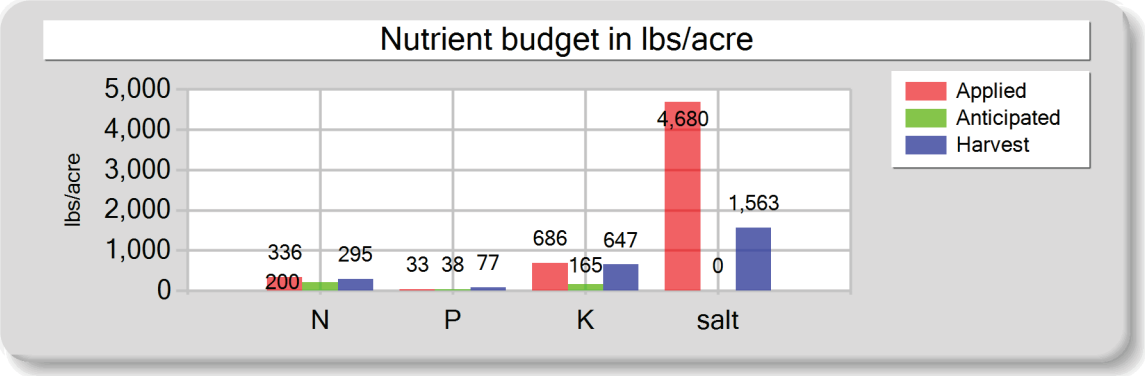
Fresh water applied	
7,290,000.00	*gallons*
268.47	*acre-inches*
9.59	*inches/acre*

Process wastewater applied	
1,350,000.00	*gallons*
49.72	*acre-inches*
1.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

August 4 - 05/22/2023: Corn, silage

Field name: August 4 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	328.96	33.42	685.70	2,836.96
Fresh water	0.00	0.00	0.00	1,842.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.96	33.42	685.70	4,679.93
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	295.29	77.23	647.36	1,562.61
Nutrient balance	40.67	-43.81	38.34	3,117.33
Applied to removed ratio	1.14	0.43	1.06	2.99

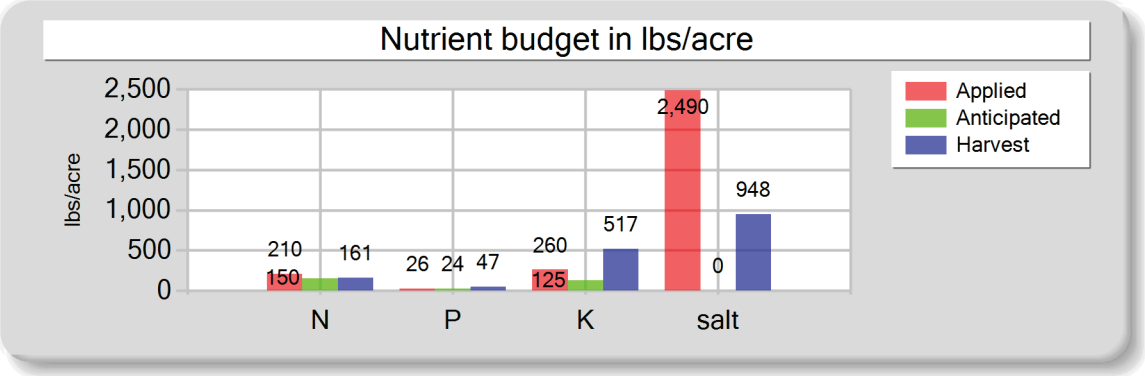
Fresh water applied	
31,711,500.00	*gallons*
1,167.83	*acre-inches*
41.71	*inches/acre*

Process wastewater applied	
2,227,500.00	*gallons*
82.03	*acre-inches*
2.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 1 - 11/01/2022: Oats, silage-soft dough

Field name: Lander 1 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.50	25.62	260.43	1,970.57
Fresh water	0.00	0.00	0.00	519.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.50	25.62	260.43	2,489.57
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	161.20	47.45	516.75	947.70
Nutrient balance	48.30	-21.83	-256.32	1,541.87
Applied to removed ratio	1.30	0.54	0.50	2.63

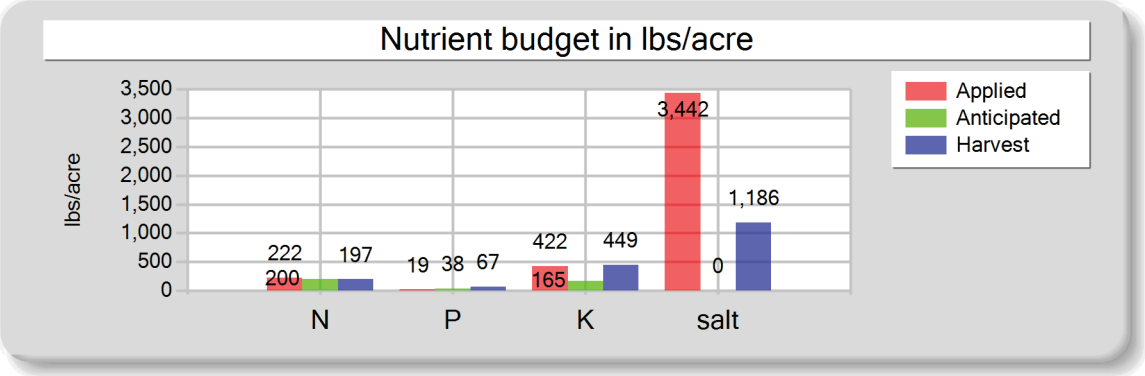
Fresh water applied	
10,206,000.00	*gallons*
375.85	*acre-inches*
11.75	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 1 - 05/24/2023: Corn, silage

Field name: Lander 1 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.54	18.82	422.11	1,680.71
Fresh water	0.00	0.00	0.00	1,760.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.54	18.82	422.11	3,441.60
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	197.26	67.41	449.04	1,186.39
Nutrient balance	24.28	-48.58	-26.92	2,255.21
Applied to removed ratio	1.12	0.28	0.94	2.90

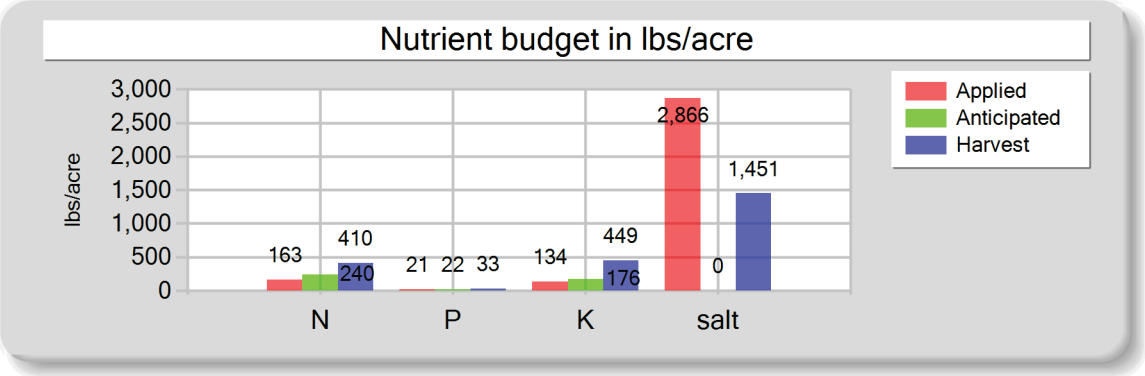
Fresh water applied	
34,627,500.00	*gallons*
1,275.21	*acre-inches*
39.85	*inches/acre*

Process wastewater applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 10 - 01/01/2021: Alfalfa, hay

Field name: Lander 10 Crop: Alfalfa, hay Plant date: 01/01/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	148.96	21.07	133.52	888.74
Fresh water	0.00	0.00	0.00	1,977.14
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	162.96	21.07	133.52	2,865.88
Anticipated crop nutrient removal	240.00	21.60	176.00	0.00
Actual crop nutrient removal	410.20	33.32	449.40	1,450.69
Nutrient balance	-247.24	-12.25	-315.88	1,415.19
Applied to removed ratio	0.40	0.63	0.30	1.98

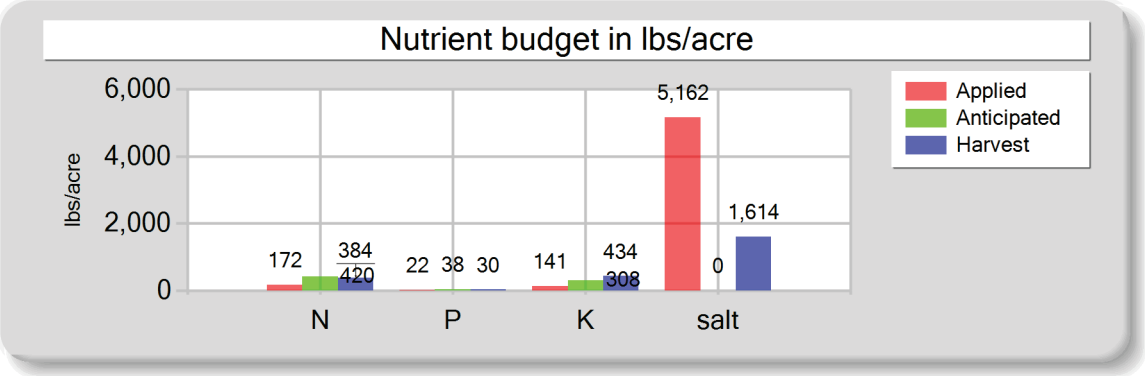
Fresh water applied	
21,870,000.00	*gallons*
805.40	*acre-inches*
44.74	*inches/acre*

Process wastewater applied	
450,000.00	*gallons*
16.57	*acre-inches*
0.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 11 - 01/01/2018: Alfalfa, hay

Field name: Lander 11 Crop: Alfalfa, hay Plant date: 01/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	157.72	22.31	141.37	941.02
Fresh water	0.00	0.00	0.00	4,221.44
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	171.72	22.31	141.37	5,162.46
Anticipated crop nutrient removal	420.00	37.80	308.00	0.00
Actual crop nutrient removal	384.04	30.14	433.59	1,613.91
Nutrient balance	-212.31	-7.83	-292.21	3,548.55
Applied to removed ratio	0.45	0.74	0.33	3.20

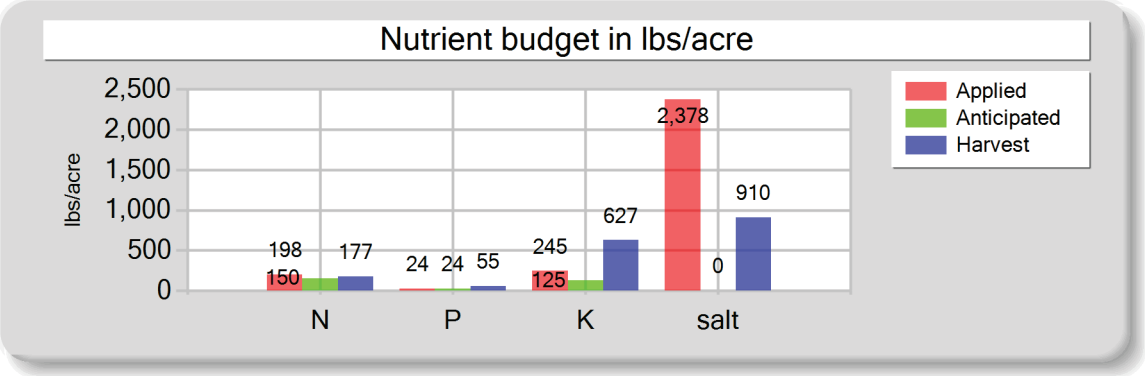
Fresh water applied	
44,101,000.00	*gallons*
1,624.09	*acre-inches*
95.53	*inches/acre*

Process wastewater applied	
450,000.00	*gallons*
16.57	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 2 & 3 - 11/01/2022: Oats, silage-soft dough

Field name: Lander 2 & 3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.59	24.11	245.11	1,854.65
Fresh water	0.00	0.00	0.00	523.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.59	24.11	245.11	2,378.01
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	176.75	55.06	626.82	909.85
Nutrient balance	20.84	-30.95	-381.72	1,468.16
Applied to removed ratio	1.12	0.44	0.39	2.61

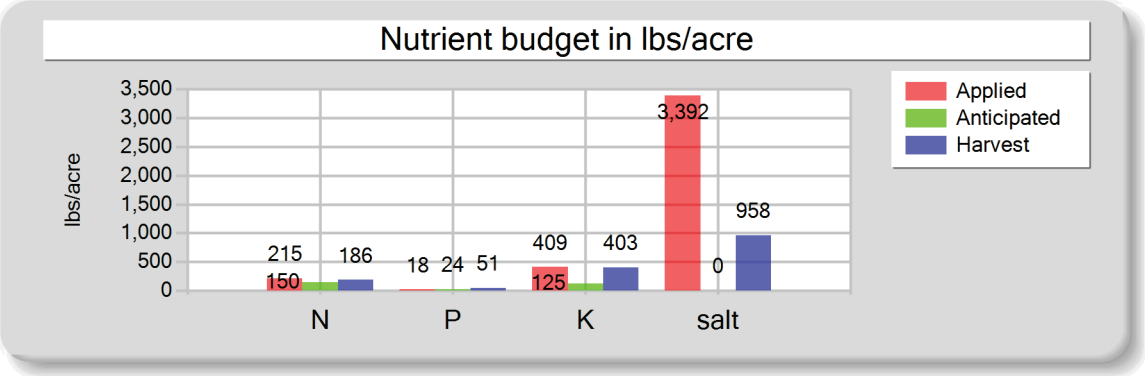
Fresh water applied	
5,467,500.00	*gallons*
201.35	*acre-inches*
11.84	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 2 & 3 - 05/24/2023: Corn, silage

Field name: Lander 2 & 3 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	208.04	18.25	409.32	1,629.78
Fresh water	0.00	0.00	0.00	1,761.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.04	18.25	409.32	3,391.76
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	185.82	50.51	402.62	957.67
Nutrient balance	29.22	-32.26	6.70	2,434.09
Applied to removed ratio	1.16	0.36	1.02	3.54

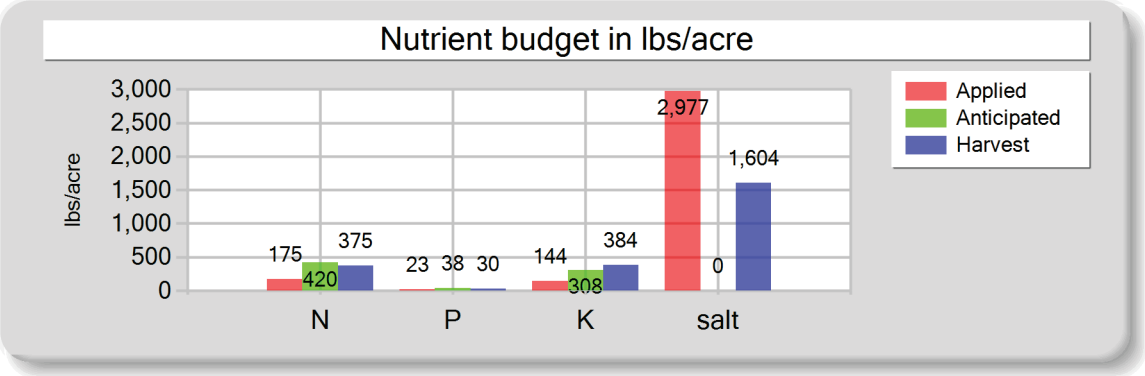
Fresh water applied	
18,407,250.00	*gallons*
677.88	*acre-inches*
39.88	*inches/acre*

Process wastewater applied	
765,000.00	*gallons*
28.17	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 4 & 6 - 01/01/2018: Alfalfa, hay

Field name: Lander 4 & 6 Crop: Alfalfa, hay Plant date: 01/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	160.87	22.76	144.20	959.84
Fresh water	0.00	0.00	0.00	2,016.68
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	174.87	22.76	144.20	2,976.52
Anticipated crop nutrient removal	420.00	37.80	308.00	0.00
Actual crop nutrient removal	375.00	30.25	383.75	1,604.25
Nutrient balance	-200.13	-7.49	-239.55	1,372.27
Applied to removed ratio	0.47	0.75	0.38	1.86

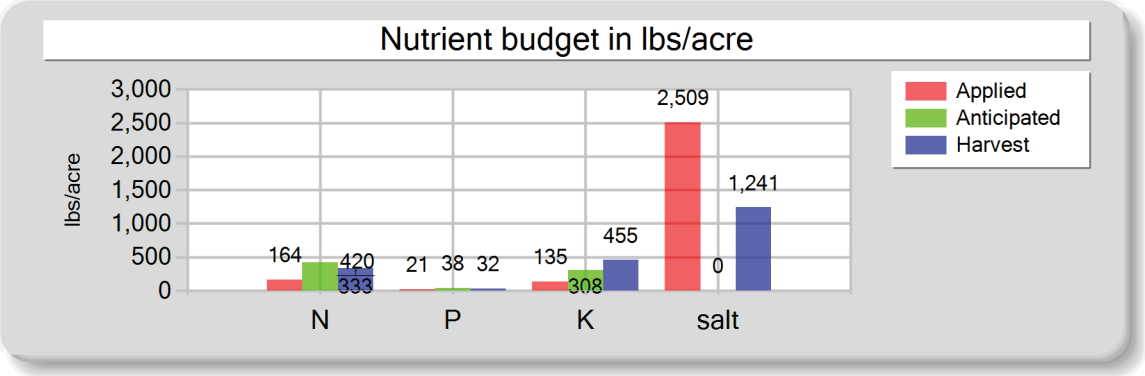
Fresh water applied	
24,786,000.00	*gallons*
912.78	*acre-inches*
45.64	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 5 & 7 - 01/01/2018: Alfalfa, hay

Field name: Lander 5 & 7 Crop: Alfalfa, hay Plant date: 01/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.15	21.24	134.59	895.85
Fresh water	0.00	0.00	0.00	1,613.35
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	164.15	21.24	134.59	2,509.20
Anticipated crop nutrient removal	420.00	37.80	308.00	0.00
Actual crop nutrient removal	332.64	32.03	454.72	1,240.51
Nutrient balance	-168.49	-10.79	-320.13	1,268.69
Applied to removed ratio	0.49	0.66	0.30	2.02

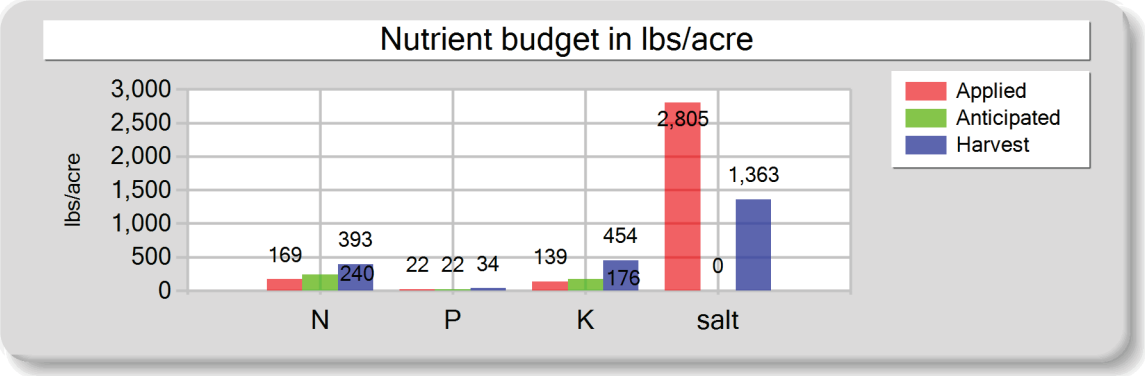
Fresh water applied	
24,786,000.00	*gallons*
912.78	*acre-inches*
36.51	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 8 - 01/01/2021: Alfalfa, hay

Field name: Lander 8 Crop: Alfalfa, hay Plant date: 01/01/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.69	21.88	138.66	922.92
Fresh water	0.00	0.00	0.00	1,882.08
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	168.69	21.88	138.66	2,805.01
Anticipated crop nutrient removal	240.00	21.60	176.00	0.00
Actual crop nutrient removal	393.08	34.33	453.65	1,362.74
Nutrient balance	-224.39	-12.45	-315.00	1,442.27
Applied to removed ratio	0.43	0.64	0.31	2.06

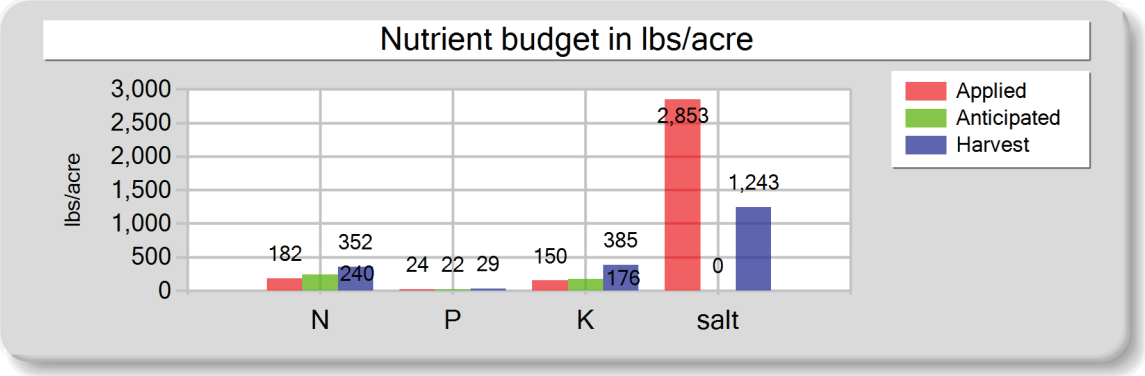
Fresh water applied	
30,071,250.00	*gallons*
1,107.42	*acre-inches*
42.59	*inches/acre*

Process wastewater applied	
675,000.00	*gallons*
24.86	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lander 9 - 01/01/2021: Alfalfa, hay

Field name: Lander 9 Crop: Alfalfa, hay Plant date: 01/01/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	167.58	23.71	150.21	999.84
Fresh water	0.00	0.00	0.00	1,853.57
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	181.58	23.71	150.21	2,853.40
Anticipated crop nutrient removal	240.00	21.60	176.00	0.00
Actual crop nutrient removal	351.60	28.56	385.20	1,243.45
Nutrient balance	-170.02	-4.85	-234.99	1,609.95
Applied to removed ratio	0.52	0.83	0.39	2.29

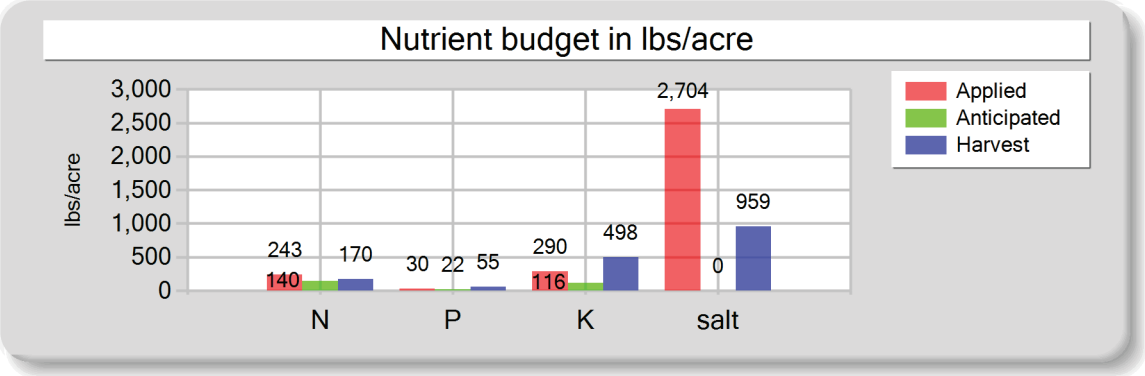
Fresh water applied	
4,556,250.00	*gallons*
167.79	*acre-inches*
41.95	*inches/acre*

Process wastewater applied	
112,500.00	*gallons*
4.14	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 1 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 1 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	236.02	30.36	290.47	2,170.53
Fresh water	0.00	0.00	0.00	533.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.02	30.36	290.47	2,704.36
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	169.96	55.16	498.40	958.66
Nutrient balance	73.06	-24.80	-207.93	1,745.70
Applied to removed ratio	1.43	0.55	0.58	2.82

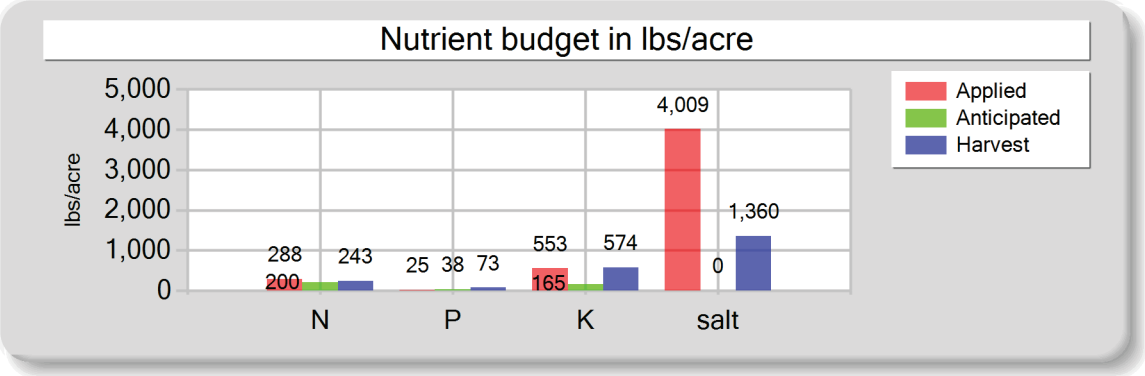
Fresh water applied	
3,280,500.00	*gallons*
120.81	*acre-inches*
12.08	*inches/acre*

Process wastewater applied	
450,000.00	*gallons*
16.57	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 1 - 05/23/2023: Corn, silage

Field name: Oslo 1 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.86	24.64	552.59	2,200.20
Fresh water	0.00	0.00	0.00	1,809.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.86	24.64	552.59	4,009.28
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	243.02	73.08	574.20	1,359.81
Nutrient balance	44.84	-48.44	-21.61	2,649.47
Applied to removed ratio	1.18	0.34	0.96	2.95

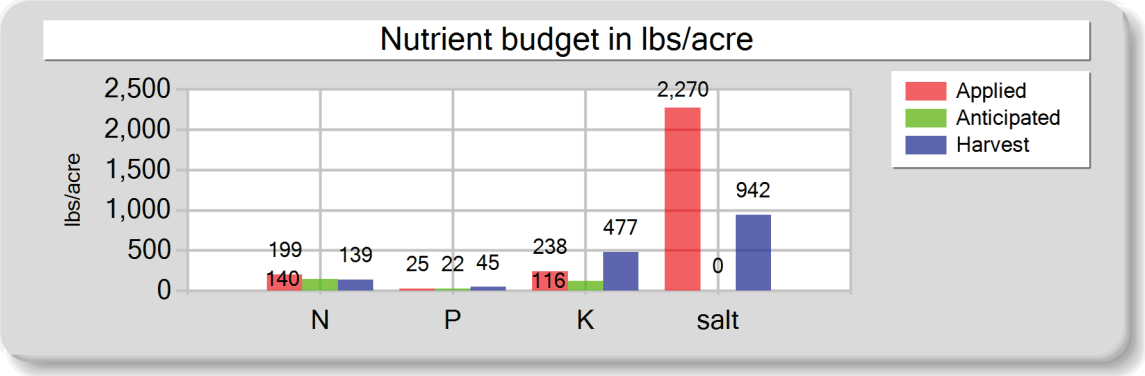
Fresh water applied	
11,117,250.00	*gallons*
409.41	*acre-inches*
40.94	*inches/acre*

Process wastewater applied	
607,500.00	*gallons*
22.37	*acre-inches*
2.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 2 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 2 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.70	24.57	238.28	1,785.72
Fresh water	0.00	0.00	0.00	483.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.70	24.57	238.28	2,269.60
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	139.26	44.62	477.47	942.43
Nutrient balance	59.43	-20.05	-239.19	1,327.17
Applied to removed ratio	1.43	0.55	0.50	2.41

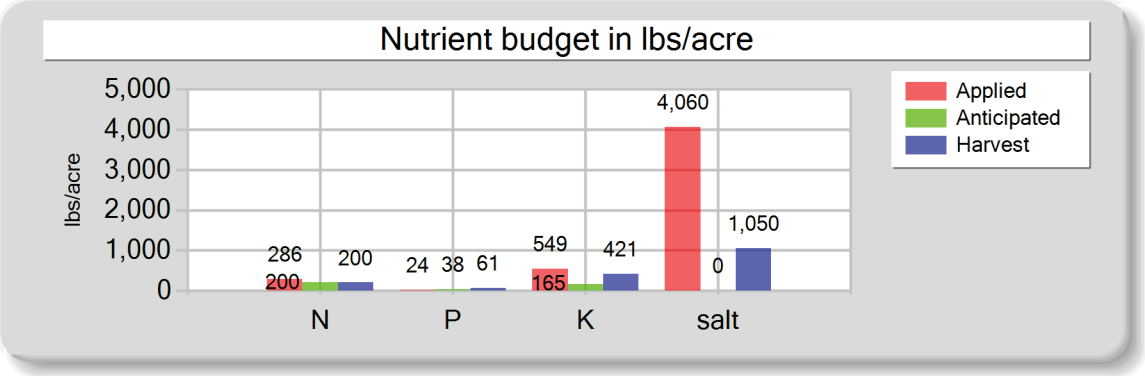
Fresh water applied	
11,299,500.00	*gallons*
416.12	*acre-inches*
10.95	*inches/acre*

Process wastewater applied	
1,395,000.00	*gallons*
51.37	*acre-inches*
1.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 2 - 05/23/2023: Corn, silage

Field name: Oslo 2 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	279.21	24.49	549.35	2,187.33
Fresh water	0.00	0.00	0.00	1,873.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.21	24.49	549.35	4,060.41
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	200.46	61.01	420.68	1,049.60
Nutrient balance	85.75	-36.52	128.67	3,010.81
Applied to removed ratio	1.43	0.40	1.31	3.87

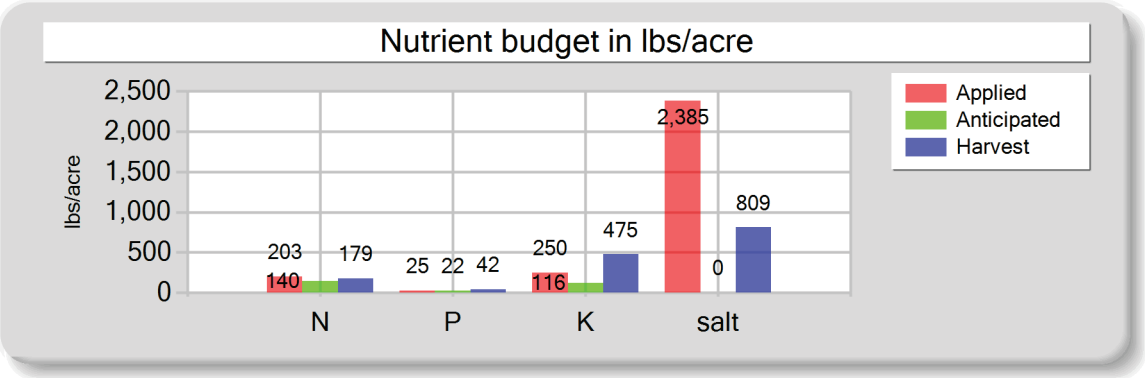
Fresh water applied	
43,740,000.00	*gallons*
1,610.80	*acre-inches*
42.39	*inches/acre*

Process wastewater applied	
2,295,000.00	*gallons*
84.52	*acre-inches*
2.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 3 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 3 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.13	24.91	249.78	1,884.90
Fresh water	0.00	0.00	0.00	500.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.13	24.91	249.78	2,385.36
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	178.65	41.63	474.75	809.19
Nutrient balance	24.48	-16.72	-224.97	1,576.17
Applied to removed ratio	1.14	0.60	0.53	2.95

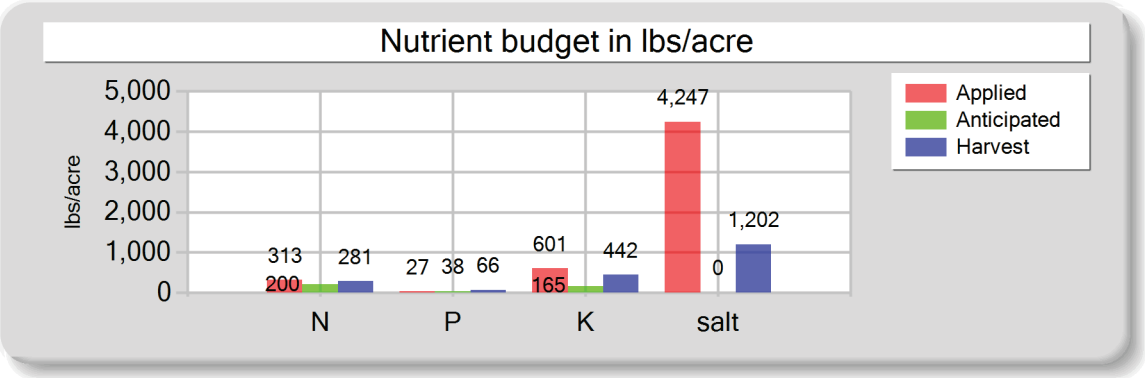
Fresh water applied	
4,920,750.00	*gallons*
181.21	*acre-inches*
11.33	*inches/acre*

Process wastewater applied	
607,500.00	*gallons*
22.37	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 3 - 05/23/2023: Corn, silage

Field name: Oslo 3 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	305.56	26.81	601.19	2,393.74
Fresh water	0.00	0.00	0.00	1,853.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.56	26.81	601.19	4,247.31
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	280.74	66.26	441.75	1,201.91
Nutrient balance	31.82	-39.46	159.44	3,045.40
Applied to removed ratio	1.11	0.40	1.36	3.53

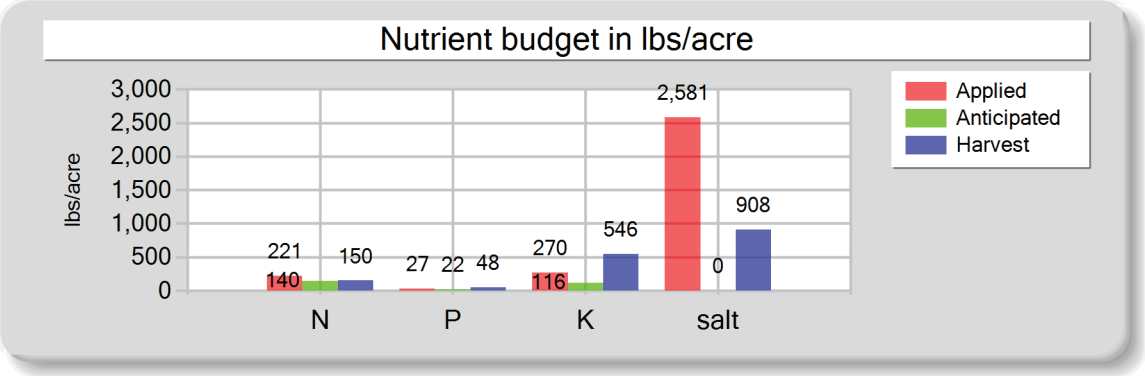
Fresh water applied	
18,225,000.00	*gallons*
671.16	*acre-inches*
41.95	*inches/acre*

Process wastewater applied	
1,057,500.00	*gallons*
38.94	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 4 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 4 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.67	27.20	270.44	2,037.22
Fresh water	0.00	0.00	0.00	543.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.67	27.20	270.44	2,580.94
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	149.85	47.79	545.75	907.73
Nutrient balance	70.82	-20.59	-275.31	1,673.20
Applied to removed ratio	1.47	0.57	0.50	2.84

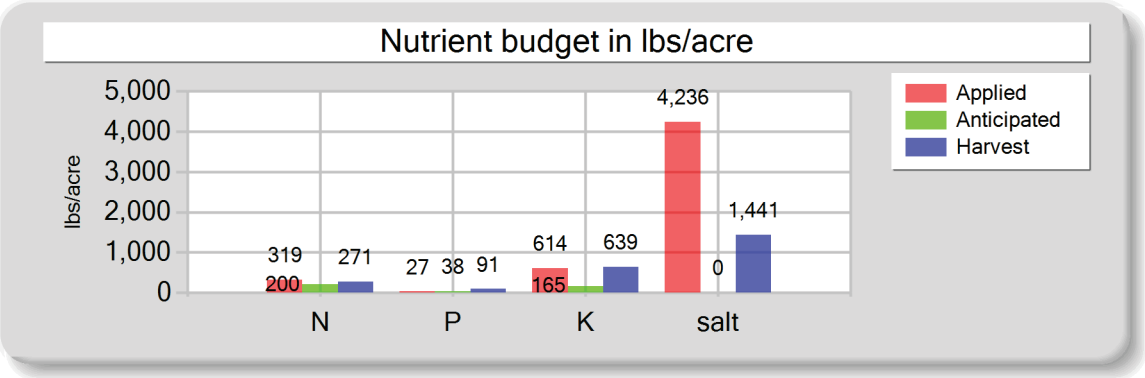
Fresh water applied	
8,019,000.00	*gallons*
295.31	*acre-inches*
12.30	*inches/acre*

Process wastewater applied	
990,000.00	*gallons*
36.46	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 4 - 05/23/2023: Corn, silage

Field name: Oslo 4 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.06	27.38	613.98	2,444.67
Fresh water	0.00	0.00	0.00	1,791.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.06	27.38	613.98	4,236.45
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	270.71	90.64	638.75	1,441.49
Nutrient balance	48.35	-63.27	-24.77	2,794.96
Applied to removed ratio	1.18	0.30	0.96	2.94

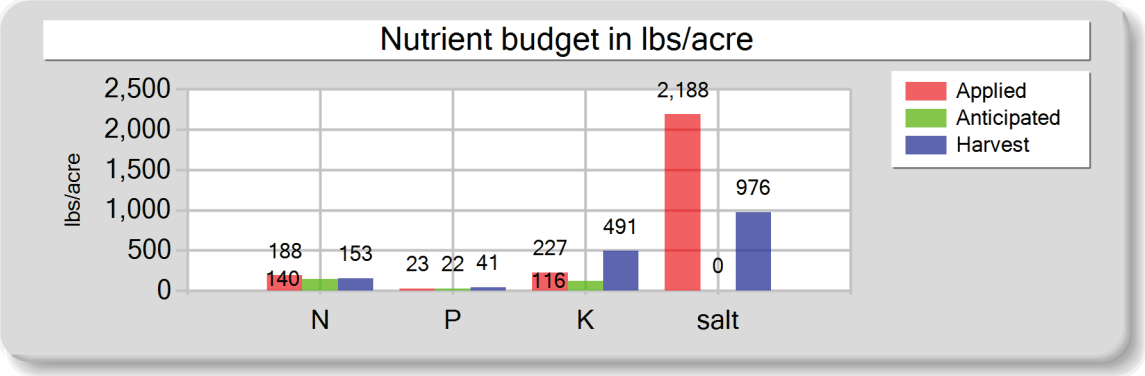
Fresh water applied	
26,426,250.00	*gallons*
973.19	*acre-inches*
40.55	*inches/acre*

Process wastewater applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 5 - 11/03/2022: Oats, silage-soft dough

Field name: Oslo 5 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	180.92	23.12	226.73	1,703.18
Fresh water	0.00	0.00	0.00	485.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.92	23.12	226.73	2,188.47
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	153.31	41.42	491.45	975.69
Nutrient balance	34.61	-18.30	-264.72	1,212.79
Applied to removed ratio	1.23	0.56	0.46	2.24

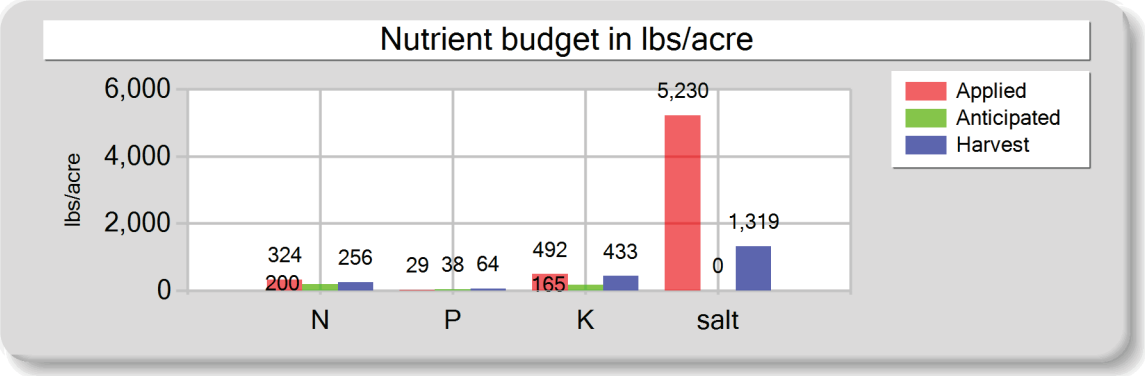
Fresh water applied	
3,280,500.00	*gallons*
120.81	*acre-inches*
10.98	*inches/acre*

Process wastewater applied	
382,500.00	*gallons*
14.09	*acre-inches*
1.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Oslo 5 - 05/23/2023: Corn, silage

Field name: Oslo 5 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	316.98	29.47	491.90	3,288.57
Fresh water	0.00	0.00	0.00	1,941.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.98	29.47	491.90	5,229.77
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	255.56	63.60	433.06	1,318.98
Nutrient balance	68.42	-34.13	58.85	3,910.78
Applied to removed ratio	1.27	0.46	1.14	3.96

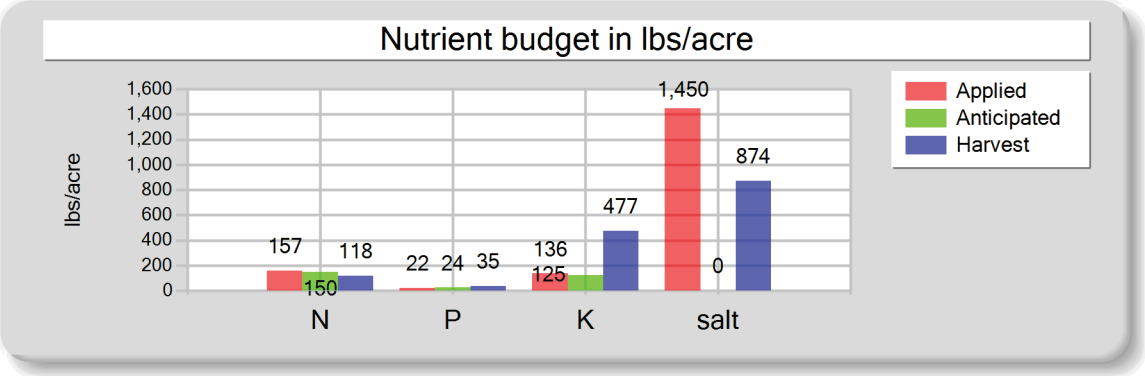
Fresh water applied	
13,122,000.00	*gallons*
483.24	*acre-inches*
43.93	*inches/acre*

Process wastewater applied	
742,500.00	*gallons*
27.34	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Verde - 11/04/2022: Oats, silage-soft dough

Field name: Verde | Crop: Oats, silage-soft dough | Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.04	21.53	136.36	1,004.74
Fresh water	0.00	0.00	0.00	444.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.04	21.53	136.36	1,449.59
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	118.13	34.65	477.00	874.35
Nutrient balance	38.92	-13.12	-340.64	575.24
Applied to removed ratio	1.33	0.62	0.29	1.66

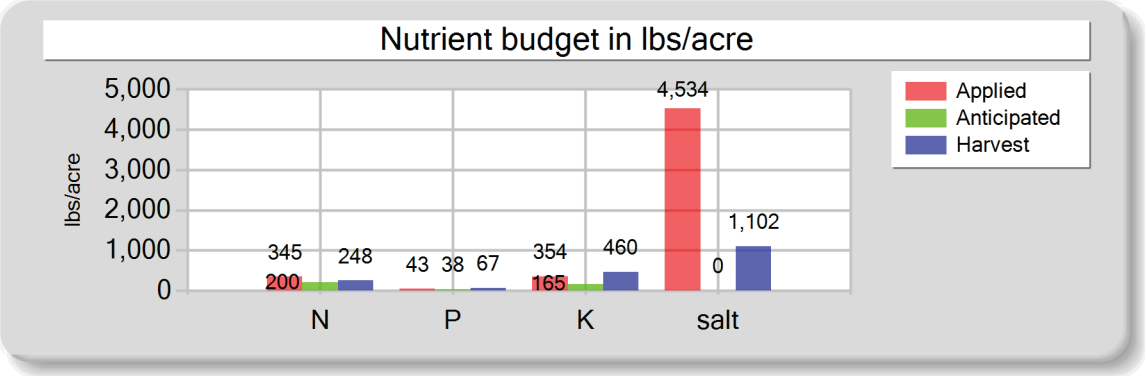
Fresh water applied	
4,374,000.00	*gallons*
161.08	*acre-inches*
10.07	*inches/acre*

Process wastewater applied	
320,000.00	*gallons*
11.78	*acre-inches*
0.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Verde - 06/12/2023: Corn, silage

Field name: Verde | Crop: Corn, silage | Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	338.05	42.79	353.93	2,513.83
Fresh water	0.00	0.00	0.00	2,020.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.05	42.79	353.93	4,534.22
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	247.92	66.73	460.04	1,101.80
Nutrient balance	97.13	-23.94	-106.11	3,432.42
Applied to removed ratio	1.39	0.64	0.77	4.12

Fresh water applied	
19,865,250.00	*gallons*
731.57	*acre-inches*
45.72	*inches/acre*

Process wastewater applied	
870,000.00	*gallons*
32.04	*acre-inches*
2.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,980.00	2,160.00	5,020.00							49.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	2,260.00	3,530.00							0.00
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,980.00	2,160.00	5,020.00							72.70
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	3,370.00	5,640.00							51.50
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,400.00	1,850.00	2,850.00	6,050.00	1,910.00	920.00	2,070.00	3,980.00		33.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	543.00	358.00	0.00	0.00	60.70	907.00								7,180.00	7,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	714.00	339.00	0.00	0.00	101.00	640.00								5,560.00	4,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	554.00	384.00	0.00	0.00	48.60	1,090.00								7,740.00	4,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	168.00	104.00	0.00	0.00	60.10	713.00	86.90	81.50	151.00	1,700.00	0.00	151.00	322.00	4,200.00	3,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	434.00	291.00	0.00	0.00	33.60	491.00								5,110.00	4,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Canal

TID Canal

Sample description: TID Canal

Sample date: 10/15/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								150.00	195
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

August 1

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			17.00				
DL			4.00				

August 2

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			22.00				
DL			4.00				

August 3

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			4.00				

August 4

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			4.00				

Lander 1

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.00				
DL			4.00				

Lander 10

Lander 10

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			4.00				

Lander 11

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			4.00				

Lander 2 & 3

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

Lander 4 & 6

Lander 4 & 6

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			19.00				
DL			4.00				

Lander 5 & 7

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			16.00				
DL			4.00				

Lander 8

S

Sample and source description: S

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			14.00				
DL			4.00				

Lander 9

Lander 9

Soil

Sample and source description: Soil

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			14.00				
DL			4.00				

Oslo 1

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			37.00				
DL			4.00				

Oslo 2

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			110.00				
DL			4.00				

Oslo 3

Oslo 3

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			4.00				

Oslo 4

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			90.00				
DL			4.00				

Oslo 5

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			77.00				
DL			4.00				

Verde

Verde

Soil

Sample and source description: Soil

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			32.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

August 1 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,290.00	2,090.00	26,100.00		12.60
DL	100.00	100.00	100.00		0.01

August 1 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,270.00	1,110.00	6,290.00		3.99
DL	100.00	100.00	100.00		0.01

August 2 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,390.00	1,960.00	22,600.00		9.88
DL	100.00	100.00	100.00		0.01

August 2 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,390.00	1,220.00	8,400.00		4.80
DL	100.00	100.00	100.00		0.01

August 3 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,480.00	1,480.00	18,100.00		11.10
DL	100.00	100.00	100.00		0.01

August 3 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,220.00	1,240.00	10,200.00		6.00
DL	100.00	100.00	100.00		0.01

August 4 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,670.00	1,620.00	18,900.00		9.53
DL	100.00	100.00	100.00		0.01

August 4 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,360.00	11,400.00		6.49
DL	100.00	100.00	100.00		0.01

Lander 1 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,960.00	1,460.00	15,900.00		9.00
DL	100.00	100.00	100.00		0.01

Lander 1 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	1,360.00	9,060.00		5.81
DL	100.00	100.00	100.00		0.01

Lander 10 - 01/01/2021: Alfalfa, hay

Alflfa

Sample and source description: Alflfa

Sample date: 12/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,300.00	2,380.00	32,100.00		11.30
DL	100.00	100.00	100.00		0.01

Lander 11 - 01/01/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 12.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,900.00	2,190.00	31,500.00		13.40
DL	100.00	100.00	100.00		0.01

Lander 2 & 3 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,260.00	1,950.00	22,200.00		10.60
DL	100.00	100.00	100.00		0.01

Lander 2 & 3 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	1,060.00	8,450.00		5.71
DL	100.00	100.00	100.00		0.01

Lander 5 & 7 - 01/01/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,700.00	2,860.00	40,600.00		12.00
DL	100.00	100.00	100.00		0.01

Lander 8 - 01/01/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,200.00	2,550.00	33,700.00		11.10
DL	100.00	100.00	100.00		0.01

Lander 9 - 01/01/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 12/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,300.00	2,380.00	32,100.00		11.30
DL	100.00	100.00	100.00		0.01

Oslo 1 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,070.00	1,970.00	17,800.00		10.60
DL	100.00	100.00	100.00		0.01

Oslo 1 - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,190.00	1,260.00	9,900.00		5.21
DL	100.00	100.00	100.00		0.01

Oslo 2 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	1,570.00	16,800.00		9.06
DL	100.00	100.00	100.00		0.01

Oslo 2 - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,450.00	1,050.00	7,240.00		4.99
DL	100.00	100.00	100.00		0.01

Oslo 3 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,940.00	1,850.00	21,100.00		11.10
DL	100.00	100.00	100.00		0.01

Oslo 3 - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,830.00	1,140.00	7,600.00		4.90
DL	100.00	100.00	100.00		0.01

Oslo 4 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,860.00	1,550.00	17,700.00		8.00
DL	100.00	100.00	100.00		0.01

Oslo 4 - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,450.00	1,490.00	10,500.00		5.41
DL	100.00	100.00	100.00		0.01

Oslo 5 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,960.00	1,340.00	15,900.00		8.72
DL	100.00	100.00	100.00		0.01

Oslo 5 - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,420.00	1,100.00	7,490.00		5.79
DL	100.00	100.00	100.00		0.01

Verde - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,570.00	1,230.00	8,480.00		4.79
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

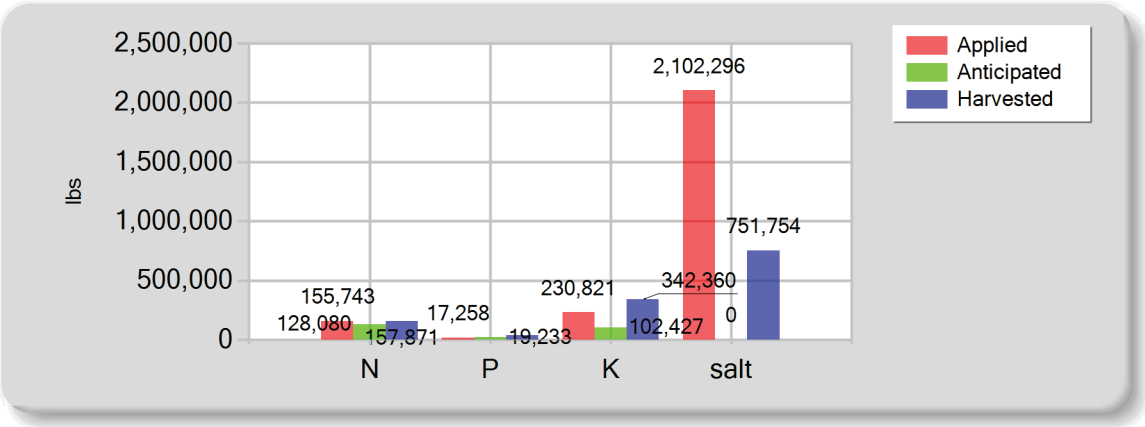
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

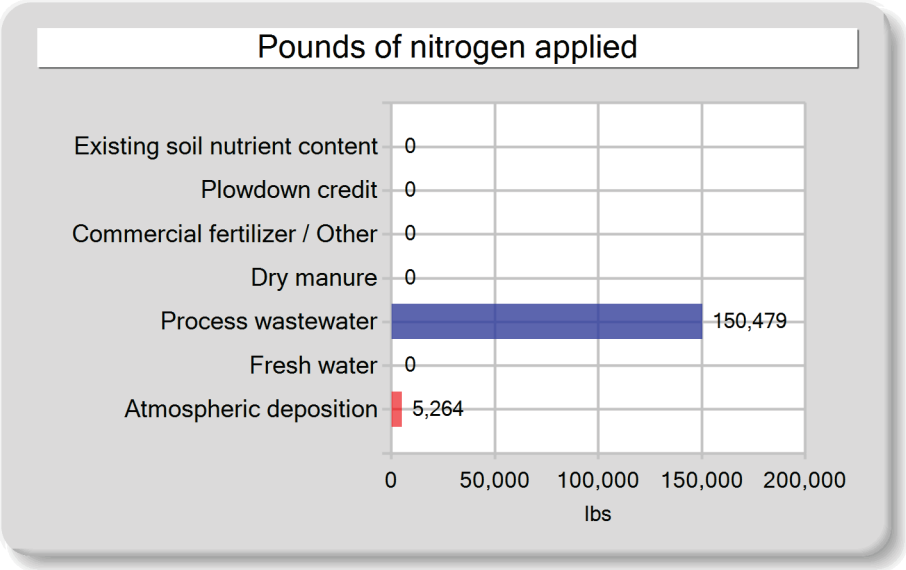
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

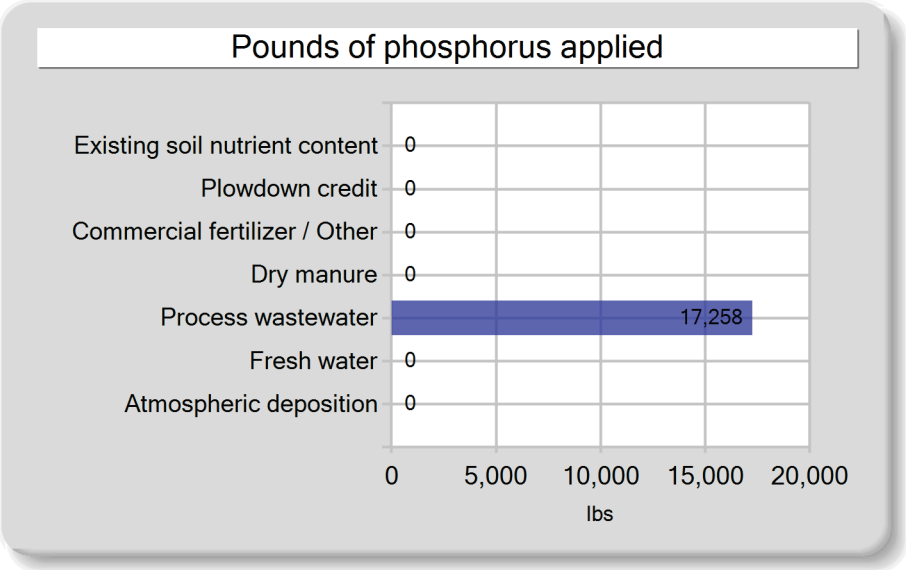
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150,479.33	17,258.12	230,820.68	1,241,949.63
Fresh water	0.00	0.00	0.00	860,346.39
Atmospheric deposition	5,264.00	0.00	0.00	0.00
Total nutrients applied	155,743.33	17,258.12	230,820.68	2,102,296.03
Anticipated crop nutrient removal	128,080.00	19,233.10	102,426.60	0.00
Actual crop nutrient removal	157,870.88	36,215.52	342,360.26	751,753.91
Nutrient balance	-2,127.55	-18,957.40	-111,539.58	1,350,542.12
Applied to removed ratio	0.99	0.48	0.67	2.80

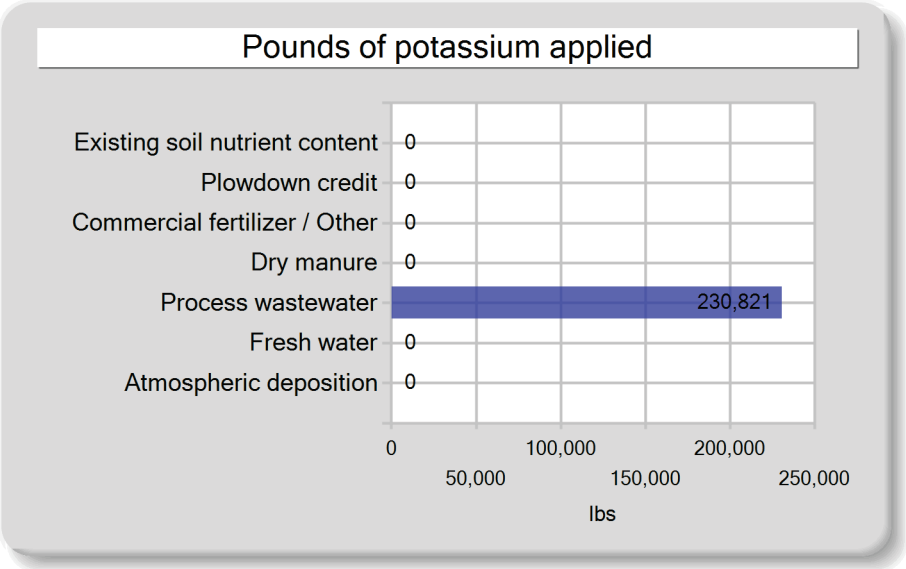
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

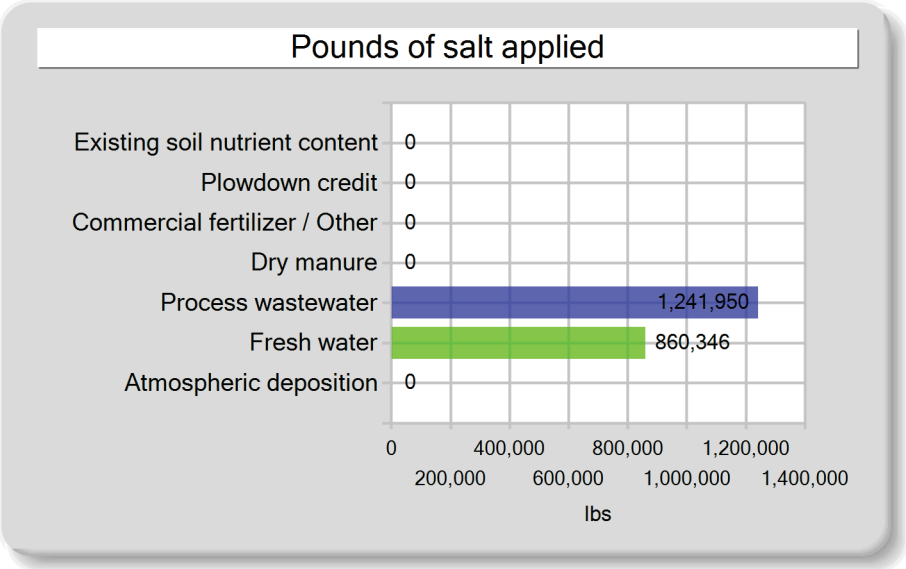


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:56 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brad Nyman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 19, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nyman

Name of Dairy Facility: Nyman Brothers Dairy

Facility Address:

Number and Street	City	County	Zip Code
20445 August AVE	Hilmar	Merced	95324

Contact Person Name and Phone Number: Brad Nyman (Name) (209) 495-8821 (Phone Number)

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Pipeline

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
20445 August AVE	Hilmar	CA	95324

Contact Person: Brad Nyman (Name) (209) 495-8821 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name: Oslo Dairy
Phone Number: (209) 495-8821

Address	City	State	Zip Code
20334 W Oslo RD	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code

Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County
0045-0014-0022-0000	Merced

Last date hauled: 08/11/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 1,102,500 gallons

Method used to determine volume of process wastewater:

Flowmeter

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:56 PDT)
Operator Signature

Jun 19, 2024
Date

Hauler Signature

Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nyman

Name of Dairy Facility: Nyman Brothers Dairy

Facility Address:

Number and Street	City	County	Zip Code
20445 August AVE	Hilmar	Merced	95324

Contact Person Name and Phone Number: Brad Nyman (Name) (209) 495-8821 (Phone Number)

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Pipeline

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
20445 August AVE	Hilmar	CA	95324

Contact Person: Brad Nyman (Name) (209) 495-8821 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Oslo Dairy (Name) (209) 495-8821 (Phone Number)

Address	City	State	Zip Code
20334 W Oslo RD	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code

Street and nearest cross street (if no address) County

Assessor's Parcel Number	Assessor's Parcel Number County
0045-0014-0035-0000	Merced

Last date hauled: 07/13/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 3,240,000 gallons

Method used to determine volume of process wastewater:

Flowmeter

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:56 PDT)
Operator Signature

Jun 19, 2024
Date

Hauler Signature

Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nyman

Name of Dairy Facility: Nyman Brothers Dairy

Facility Address:

20445 August AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Brad Nyman (209) 495-8821
Name Phone Number

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Pipeline

Address of Hauling Company/Person:

20445 August AVE	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Brad Nyman (209) 495-8821
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Oslo Dairy (209) 495-8821
Name Phone Number

20334 W Oslo RD	Hilmar	CA	95324
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

0045-0014-0028-0000 Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 05/16/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 1,485,000 gallons

Method used to determine volume of process wastewater:

Flowmeter

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

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Brad Nyman
Brad Nyman (Jun 19, 2024 09:56 PDT)

Operator Signature — Date: Jun 19, 2024

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nyman

Name of Dairy Facility: Nyman Brothers Dairy

Facility Address:

20445 August AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Brad Nyman (209) 000-0000
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons Custom Spreading

Address of Hauling Company/Person:

P.O. Box 1261	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Frank Silva (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Frank Silva (209) 632-8202
Name Phone Number

23616 Williams AVE	Hilmar	CA	95324
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

23616 Williams AVE	Hilmar	95324
Address	City	Zip Code

Merced
Street and nearest cross street (if no address) County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 05/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,890.00 tons

Manure Solids Content: 60.4 %

Method used to determine amount of manure:

Haulers Calcs

CERTIFICATION

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Brad Nyman
Brad Nyman (Jun 19, 2024 09:56 PDT)

Operator Signature — Date: Jun 19, 2024

Frank Silva

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Brad Nyman

Name of Dairy Facility: Nyman Brothers Dairy

Facility Address:

Number and Street	City	County	Zip Code
20445 August AVE	Hilmar	Merced	95324

Contact Person Name and Phone Number: Brad Nyman (Name) (209) 495-8821 (Phone Number)

PROCESS WASTEWATER HAULER INFORMATION

Name of Hauling Company/Person: Pipeline

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
20445 August AVE	Hilmar	CA	95324

Contact Person: Brad Nyman (Name) (209) 495-8821 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name: Oslo Dairy
Phone Number: (209) 495-8821

Address	City	State	Zip Code
20334 W Oslo RD	Hilmar	CA	95324

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code

Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County
0045-0014-0022-0000	Merced

Last date hauled: 02/26/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

PROCESS WASTEWATER AMOUNT HAULED

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: 1,755,000 gallons

Method used to determine volume of process wastewater:

Flowmeter

WRITTEN AGREEMENT

Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use?

[X] YES [ ] NO

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.

_____ (Operator shall provide initials here to acknowledge this requirement)

CERTIFICATION

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Brad Nyman — Jun 19, 2024

Brad Nyman (Jun 19, 2024 09:56 PDT)

Operator Signature — Date

Hauler Signature — Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Nyman Bros. Dairy
20445 August Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Milk Barn	T3240028A	Electrical Conductivity (EC)	EPA 120.1 and	1.46	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	932	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

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Compliance Analysis Report

Customer
Nyman Bros. Dairy
20445 August Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Feed Area	T3240028B	Electrical Conductivity (EC)	EPA 120.1 and	1.45	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	929	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Nyman Bros. Dairy
20445 August Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW House	T3240028C	Electrical Conductivity (EC)	EPA 120.1 and	1.89	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.91	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1210	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

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Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Nyman Bros. Dairy
20445 August Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Shop	T3240028D	Electrical Conductivity (EC)	EPA 120.1 and	1.84	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.61	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1180	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	95.2 (T3240023E)	96.4 (T3240023E)	1.25
Electrical Conductivity (EC)	100			0.217
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Nyman
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To:
F:R
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	8/24	10:00	Domestic wells	DW Milk barn		14.0' ✓					
2	✓	10:15	✓	DW Feed Area		14.0' ✓					
3		10:30		DW House		13.6' ✓					
4		10:45		DW Shop		15.2' ✓					
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	[signature]	FRAG				
2nd		DENELE ANALYTICAL, INC.	8-28-23 12:36			T3240028

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

Nyman Bros. 2023 Annual Report

Final Audit Report 2024-06-19

Created:	2024-06-19
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA6fom9RG5iqcb7vz7njsxP3tx2ze_pN3r

"Nyman Bros. 2023 Annual Report" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-06-19 - 4:40:44 PM GMT

Document emailed to nymanbrad@gmail.com for signature
2024-06-19 - 4:43:49 PM GMT

Email viewed by nymanbrad@gmail.com
2024-06-19 - 4:55:31 PM GMT

Signer nymanbrad@gmail.com entered name at signing as Brad Nyman
2024-06-19 - 4:56:32 PM GMT

Document e-signed by Brad Nyman (nymanbrad@gmail.com)
Signature Date: 2024-06-19 - 4:56:34 PM GMT - Time Source: server

Agreement completed.
2024-06-19 - 4:56:34 PM GMT